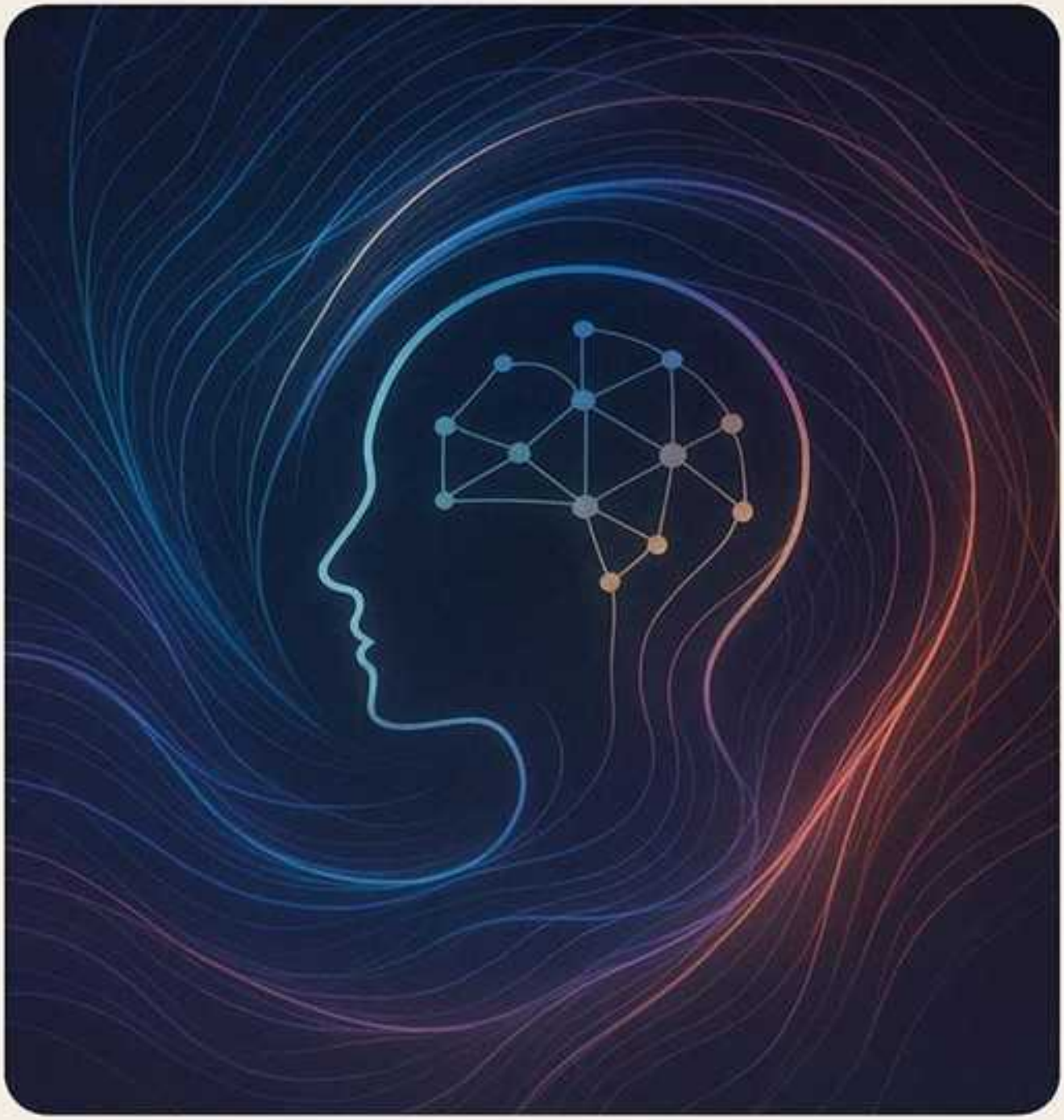
AI EDUCATION TRAINING AND DEVELOPMENT PLANNING APPLICATION

The future is not linear, it is Quantum



Developer

PIERRE WEPENER

DEVELOPER IMPORTANT NOTE

This is not a conventional Learning Management System. It is an advanced AI-powered tool designed specifically to support the planning and delivery of vocational learning programmes. By scanning the official QCTO Curriculum and Assessment Specifications, the system automatically generates a comprehensive suite of accredited learning resources, including:

- Learning material tailored to the qualification
- Roll-out plans and structured training flows
- Learning guides for facilitators and learners
- Summative and formative assessments aligned to QCTO standards
- Lecturer aids and facilitation tools
- Paraphrasing tools for academic support
- Memorandums and marking guides
- Portfolio of Evidence Templates
- Learner registration and administrative documents
- Logbooks for workplace learning
- Slide shows and presentation material
- All of this is produced in under 40 minutes, delivering what traditionally took months of preparation.

KEY ADVANTAGES

Efficiency: Entire sets of accredited material are generated in minutes. Standardization: All processes and outputs are consistent, ensuring fairness, validity, and reliability. Integrity: content remains true to QCTO curricula, SAQA qualifications, and benchmark studies.

Educator empowerment: Lecturers and facilitators can focus on subject expertise and delivery, rather than months of material preparation.

For the first time, TVET Colleges, Universities, and Technikons in South Africa will have access to standardized, locally developed learning material that meets international benchmarks. This innovation positions South African institutions to compete with the global best, while remaining firmly rooted in the national education framework.

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Figure 21: The main desk board ..... **Error! Bookmark not defined.**

ABSTRACT

This system was self-designed and built with no sponsors or funding-support. I am looking forward to doing this under these auspices of the DBE, DHET, QCTO, ETQA's or in conjunction with a BEEE Partner.

THE SYSTEMS SCAN THE QCTO CURRICULUM AND ASSESSMENT SPECIFICATION DEVELOPS A FULL OF ACCREDITED LEARNING MATERIALS ACCORDING TO TOTE QCTO CRITERIA AND GUIDELINES WITHIN 40 MINUTES.

EXECUTIVE SUMMARY (FORMAL, ACCREDITATION-FOCUSED)

ETDP AI Learning Design System *Transforming vocational education through accredited automation*

The ETDP AI Learning Design System is a pioneering platform that redefines vocational learning programme design. By scanning official QCTO curricula and assessment specifications, it generates complete suites of accredited learning resources — including learner guides, assessments, roll-out plans, logbooks, and lecturer toolkits — in under 40 minutes. This innovation compresses months of preparation into minutes, ensuring efficiency, integrity, and compliance.

KEY ADVANTAGES

- **Efficiency**: Entire sets of accredited material generated in minutes.
- **Standardization**: Consistent processes and outputs, ensuring fairness, validity, and reliability.
- **Integrity**: Content remains true to QCTO curricula, SAQA qualifications, and benchmark studies.
- **Educator empowerment**: Lecturers focus on subject expertise and delivery, not administrative preparation.

Strategic Impact

- **Audit readiness**: Built-in version control, traceability, and compliance reporting.
- **Global positioning**: Aligns with international benchmarks while rooted in South African frameworks.

- **Institutional advantage**: Enables TVET Colleges, Universities, and Technikons to compete globally with standardized, locally developed material.

CONCLUSION

The ETDP AI Learning Design System is more than a tool — it is an infrastructure for educational transformation. It empowers institutions to deliver accredited, compliant, and globally competitive programmes while restoring focus to pedagogy and learner success.

The Education and Training Learning Program Design System is an integrated digital platform for designing, organising, generating, reviewing, and exporting accredited learning programmes. The system supports the full curriculum development journey, from qualification setup and Quality Council curriculum uploads through to lesson planning, learner registration, work experience records, assessment material, learning schedules, workbooks, memoranda, progress reports, slide decks, flow diagrams, and final export packs.

At its centre, the system helps training providers turn formal curriculum and assessment specifications into practical teaching, learning, and assessment documents. It keeps the qualification structure organised by phases, subjects, topics, outcomes, lesson plan content, and learning material outputs. It also provides guided menus, workflow checks, preview screens, export functions, upload templates, diagnostics, and AI-assisted support so that curriculum development can be completed in a more consistent, traceable, and efficient way.

The platform is designed for education and training environments where accuracy, version control, audit readiness, and repeatable document generation are important. Users can upload official curriculum documents, import structured templates, review mapped content, generate draft learning and assessment material with the configured AI specialist service, and then download or save the resulting documents for moderation, printing, or institutional use. Although the system automates much of the drafting and formatting work, final academic review and approval remain the responsibility of the training provider.

MAIN MENU AND SYSTEM FUNCTIONS

Figure 1: Main Menu\ Dashboard and Workflow

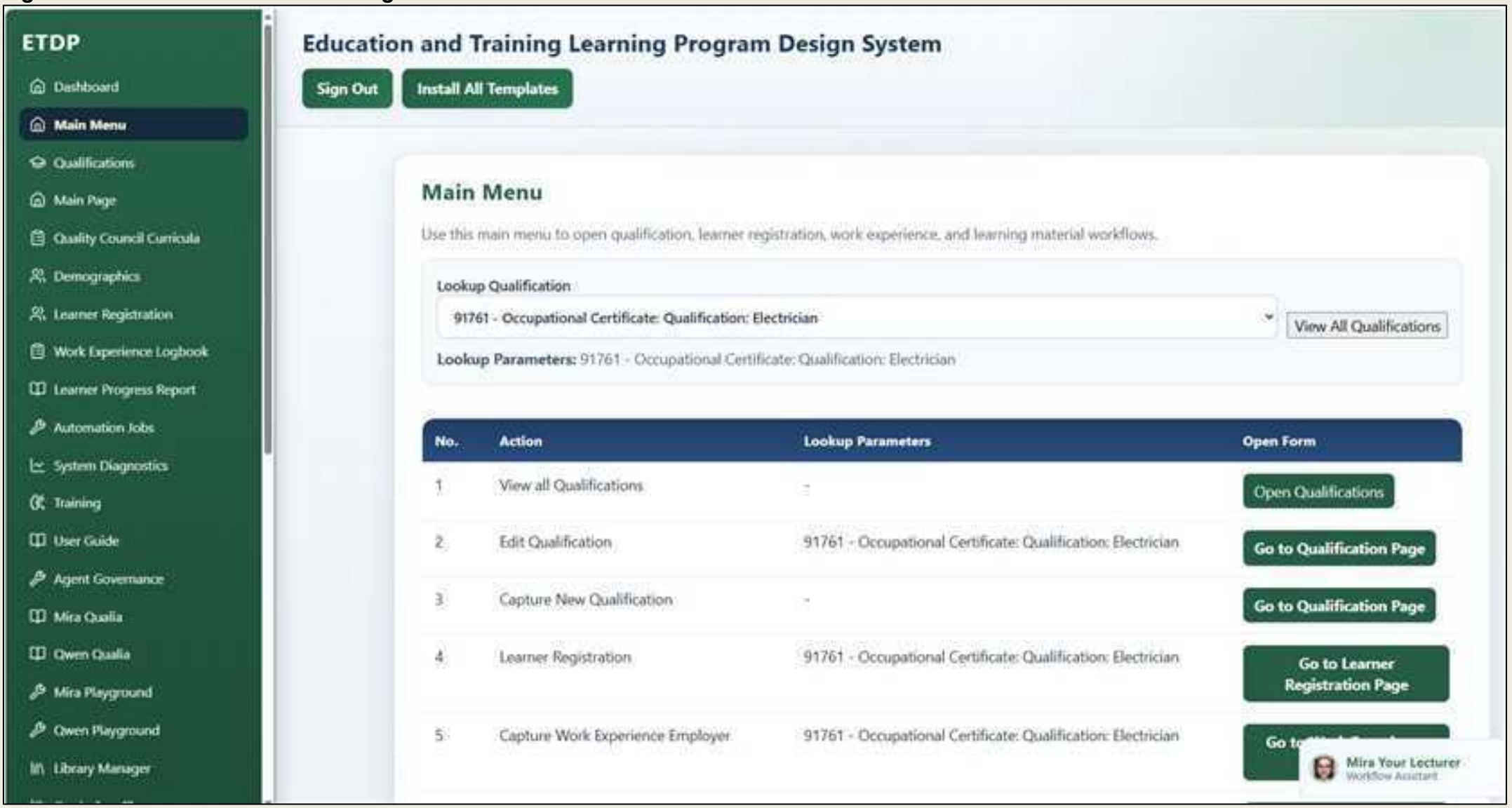
The Dashboard is the main working area of the system. It gives the user a step-by-step curriculum development workflow and a left-hand navigation menu. It helps users follow the correct order: select or capture a qualification, upload official curriculum documents, define curriculum structure, prepare lesson plan content, generate learning material, and review/export final outputs. The Main Menu is the central starting point. From here, the user can select a qualification, view all qualifications, edit an existing qualification, capture a new qualification, open learner registration, open work experience logbooks, go to the learning material dashboard, or open the training and continuous learning area.



THE QUALIFICATIONS AND MAIN PAGE

The Qualifications and Main Page functions are used to capture and maintain the qualification record. This includes qualification number, description, accreditation information, institutional details, CESM field, logo information, and related qualification metadata. This information is reused throughout the document generated.

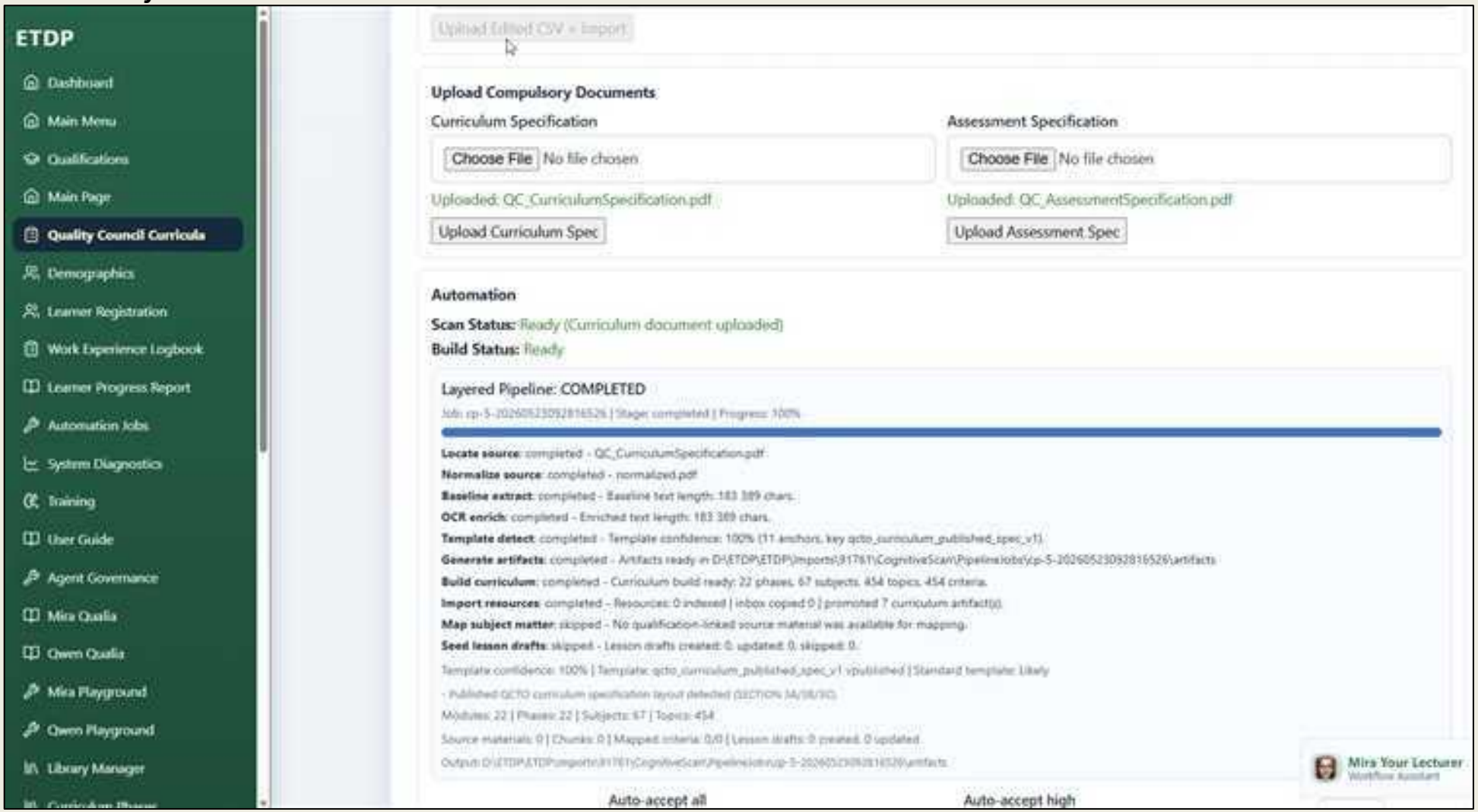
Figure 2: Qualifications and Main Page



THE QUALITY COUNCIL CURRICULA SECTION

The Quality Council Curricula section is the first major workflow gate. It allows users to upload or import compulsory curriculum and assessment specifications, import documents from a shared QCTO-style library, run cognitive scans, review curriculum mappings, download scan results, upload edited mapping files, and manage standards references such as SANS-related mapping reviews.

Figure 3 The Quality Council Curricula section

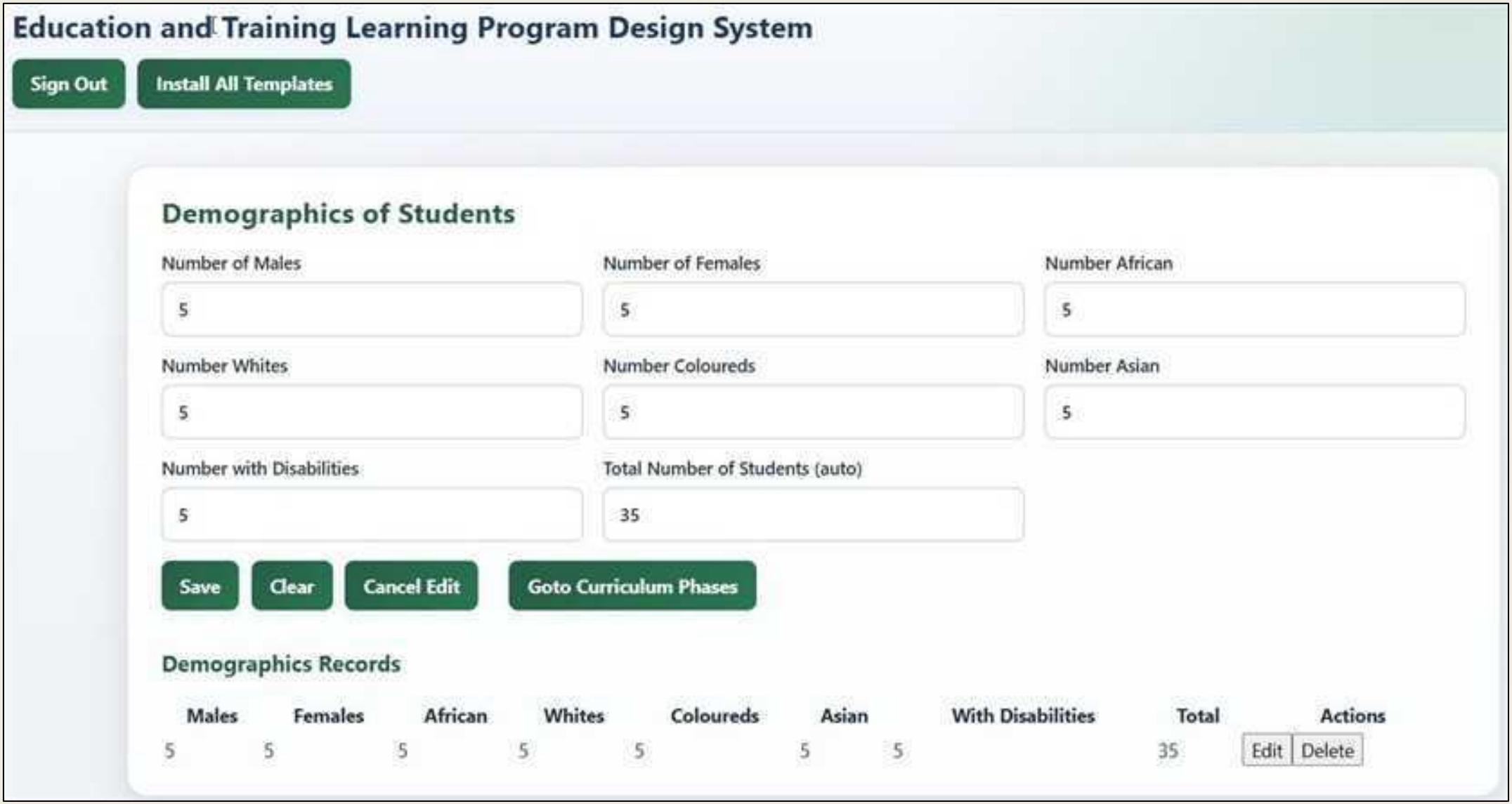


THE DEMOGRAPHICS SECTION

The Demographics section records learner and enrolment information linked to the qualification. It supports planning, reporting, and institutional records by capturing demographic details that may be required for programme documentation.

Figure 4 The learner demographics

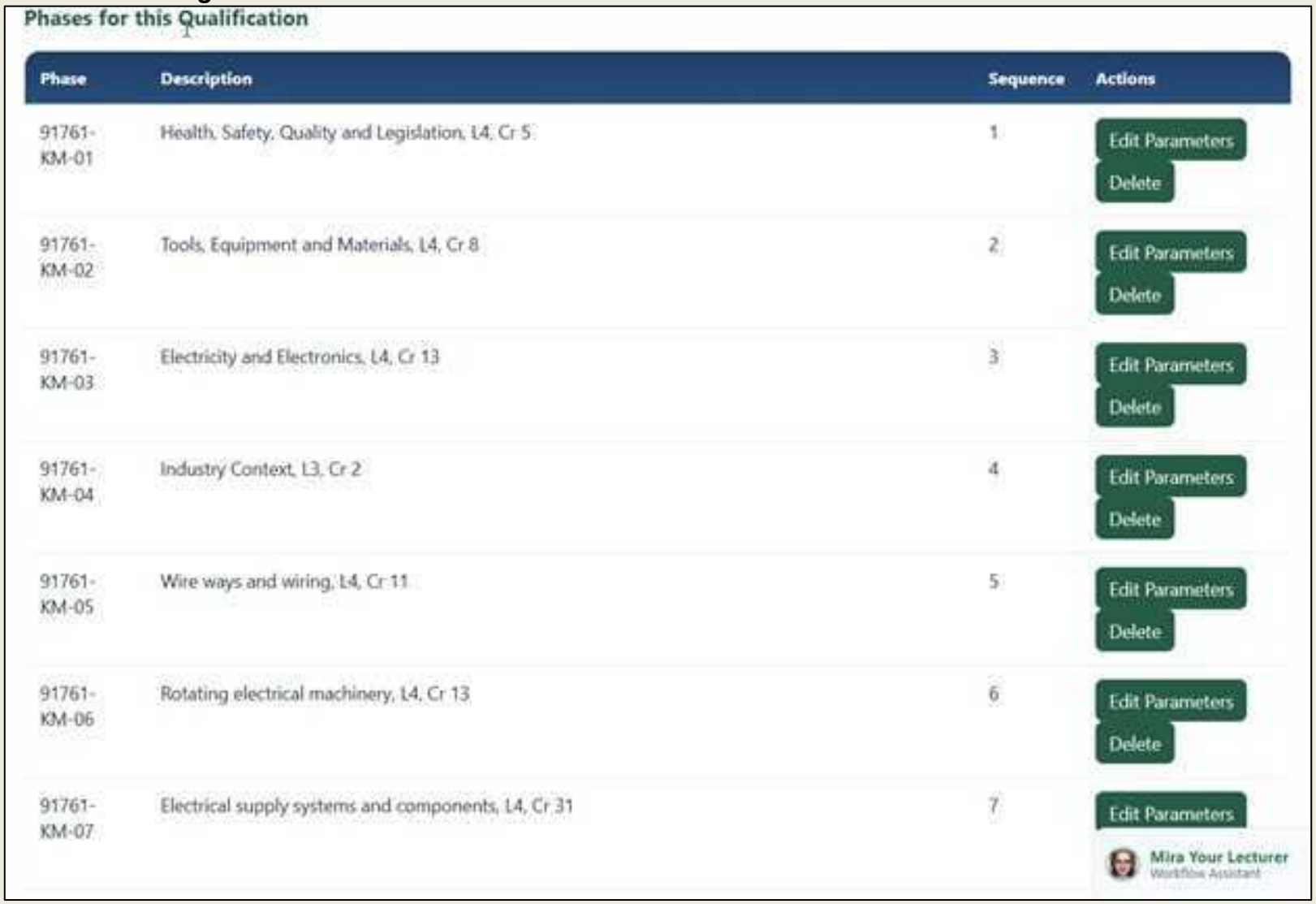



Education and Training Learning Program Design System
Sign Out
Install All Templates
Demographics of Students
Number of Males
5
Number of Females
5
Number African
5
Number Whites
5
Number Coloureds
5
Number Asian
5
Number with Disabilities
5
Total Number of Students (auto)
35
Save
Clear
Cancel Edit
Goto Curriculum Phases
Demographics Records
Males Females African Whites Coloureds Asian With Disabilities Total Actions
5 5 5 5 5 5 5 35 Edit Delete

THE CURRICULUM PHASES

The Curriculum Phases section defines the main structural phases of the qualification. Users can manually capture phases, download templates, upload phase templates, import built-in templates, preview imported data, and link phases to the selected qualification.

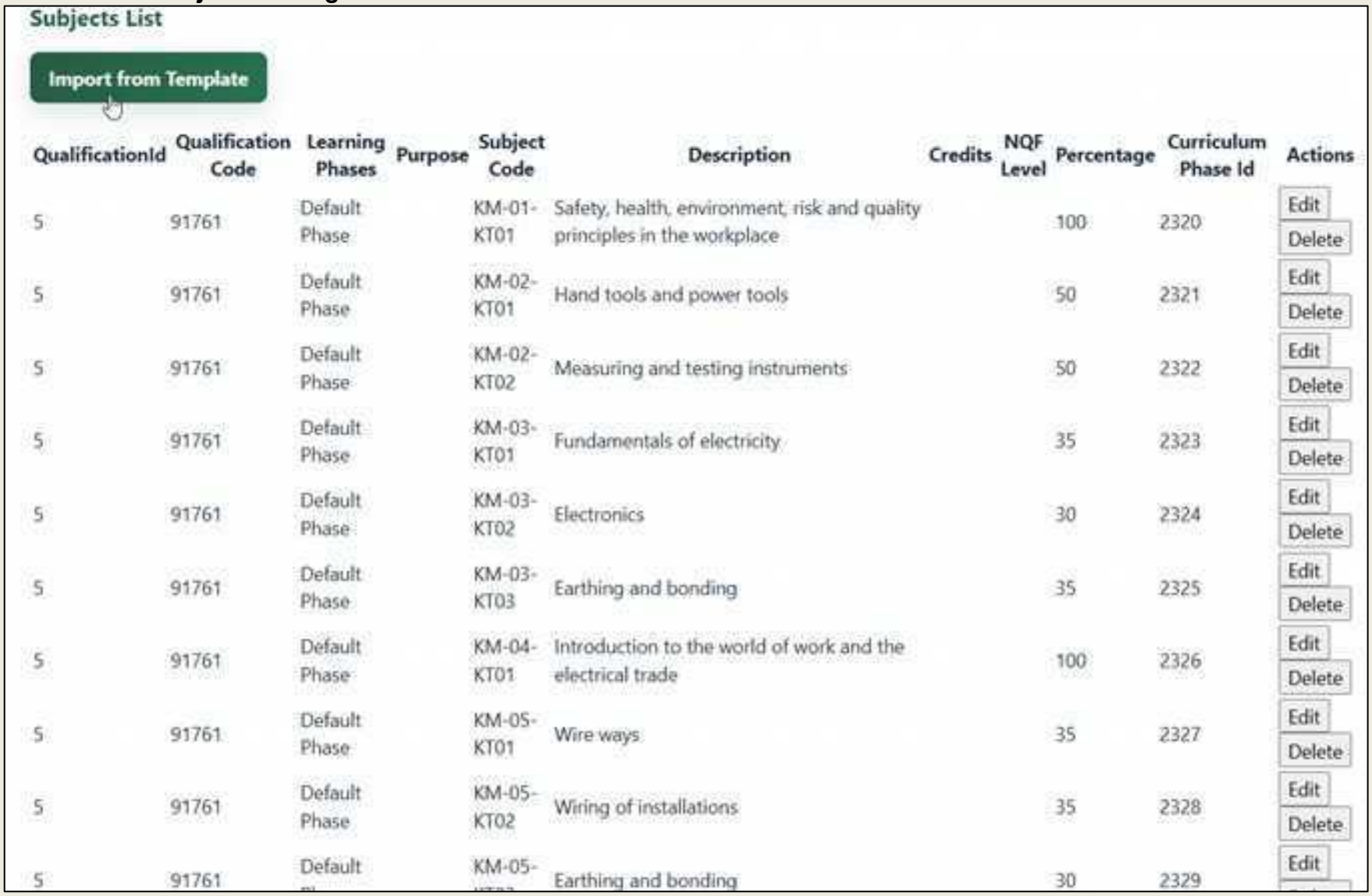
Figure 4: Curriculum Phase Manager



THE SUBJECTS SECTION CAPTURES THE SUBJECTS OR KNOWLEDGE/PRACTICAL/WORK EXPERIENCE COMPONENTS

The Subjects section captures the subjects or knowledge/practical/work experience components that belong to the curriculum phases. It supports manual entry, template downloads, Excel uploads, built-in imports, validation, preview, and subject lists.

Figure 5: The Curriculum Subjects Manager

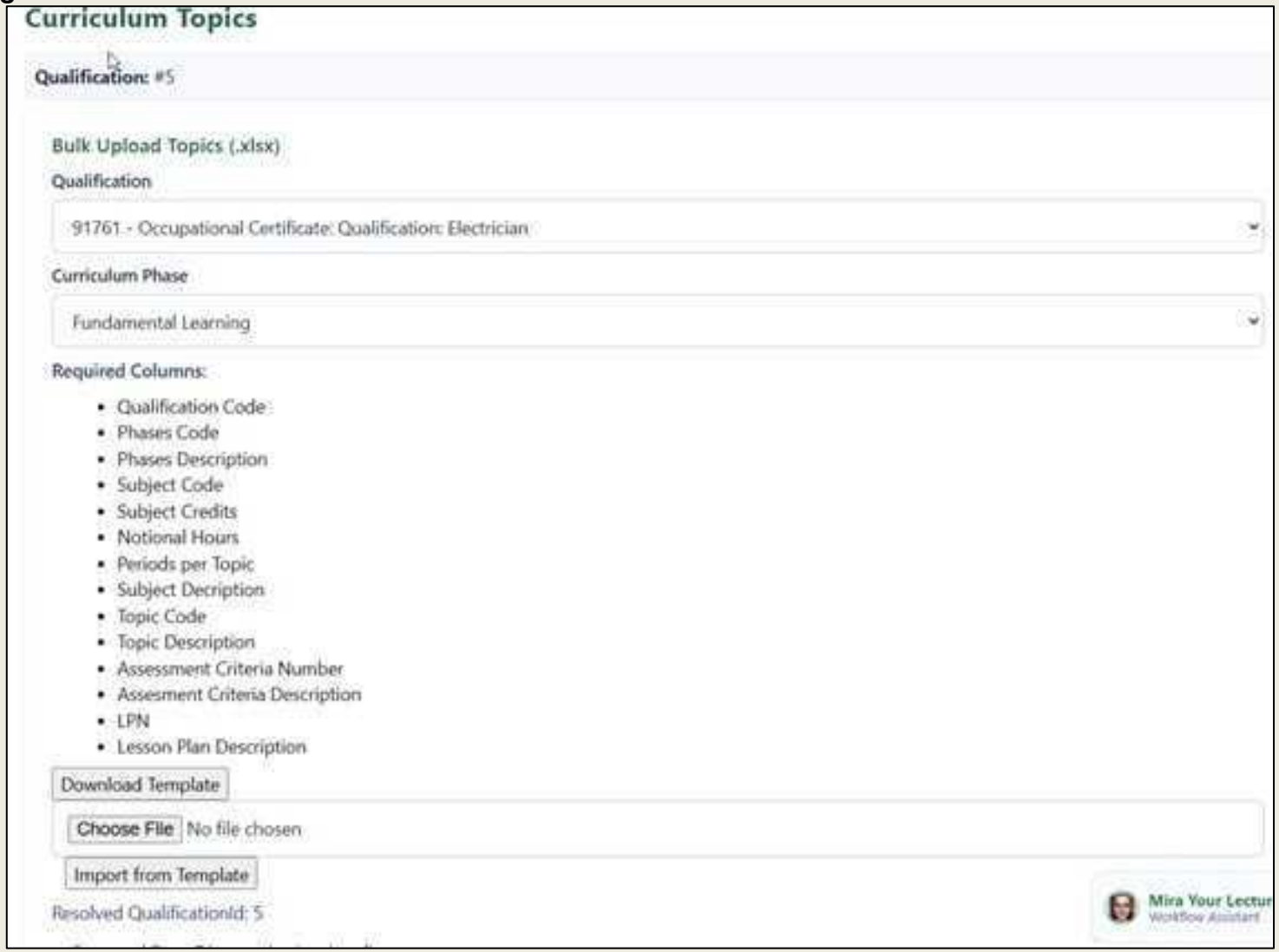


QualificationId	Qualification Code	Learning Phases	Purpose	Subject Code	Description	Credits	NQF Level	Percentage	Curriculum Phase Id	Actions
5	91761	Default Phase		KM-01-KT01	Safety, health, environment, risk and quality principles in the workplace			100	2320	Edit Delete
5	91761	Default Phase		KM-02-KT01	Hand tools and power tools			50	2321	Edit Delete
5	91761	Default Phase		KM-02-KT02	Measuring and testing instruments			50	2322	Edit Delete
5	91761	Default Phase		KM-03-KT01	Fundamentals of electricity			35	2323	Edit Delete
5	91761	Default Phase		KM-03-KT02	Electronics			30	2324	Edit Delete
5	91761	Default Phase		KM-03-KT03	Earthing and bonding			35	2325	Edit Delete
5	91761	Default Phase		KM-04-KT01	Introduction to the world of work and the electrical trade			100	2326	Edit Delete
5	91761	Default Phase		KM-05-KT01	Wire ways			35	2327	Edit Delete
5	91761	Default Phase		KM-05-KT02	Wiring of installations			35	2328	Edit Delete
5	91761	Default		KM-05-	Earthing and bonding			30	2329	Edit

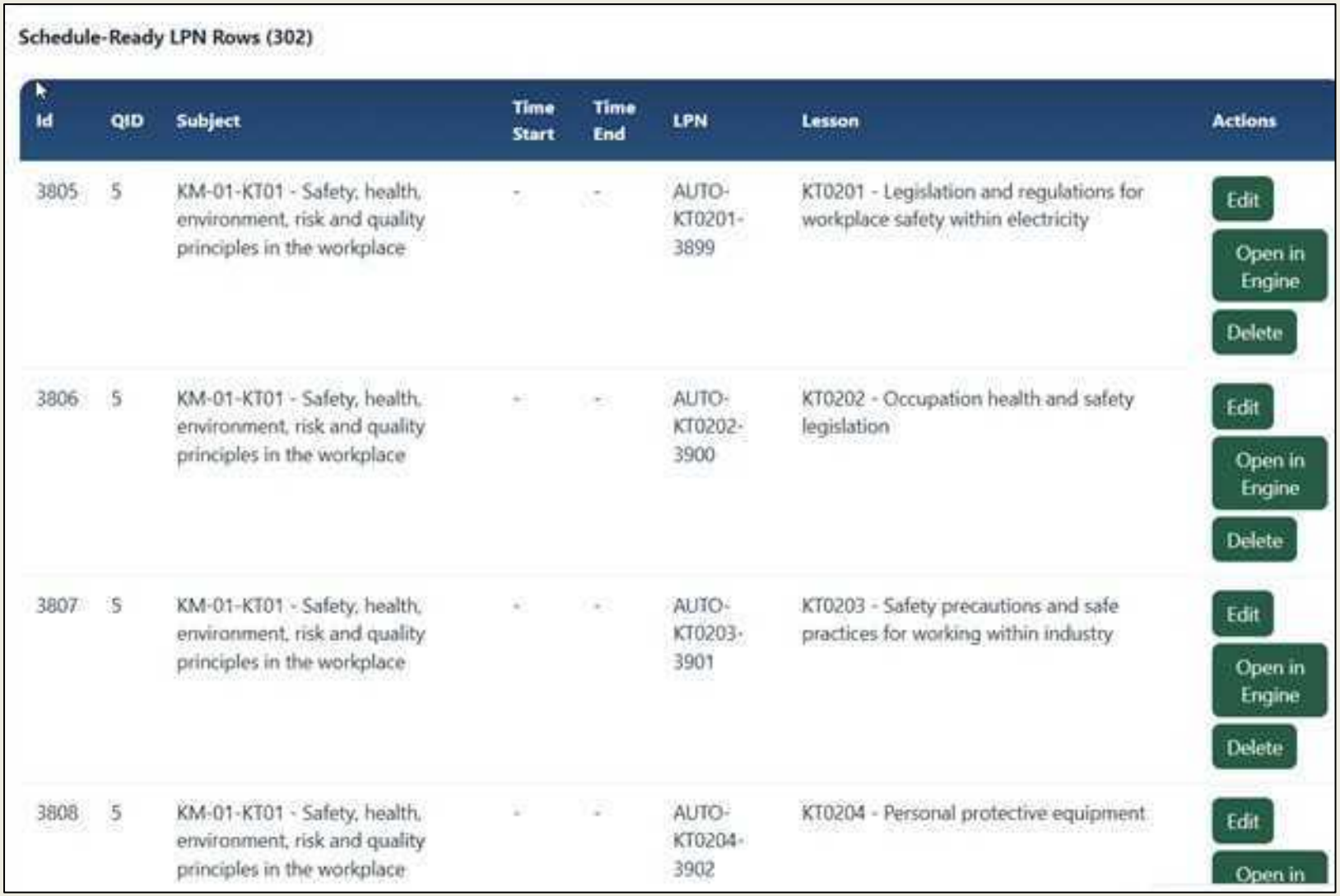
THE TOPICS SECTION

The Topics section breaks subjects into structured learning topics. These topics become the foundation for lesson plans, learner guides, workbook activities, assessments, slides, flow diagrams, and other learning material outputs. The Outcomes function supports capture and review of outcomes linked to the curriculum structure. These outcomes help align learning material and assessment content to the qualification requirements.

Figure 6: Topics Manager



THESE TOPICS BECOME THE FOUNDATION FOR LESSON PLANS



Schedule-Ready LPN Rows (302)

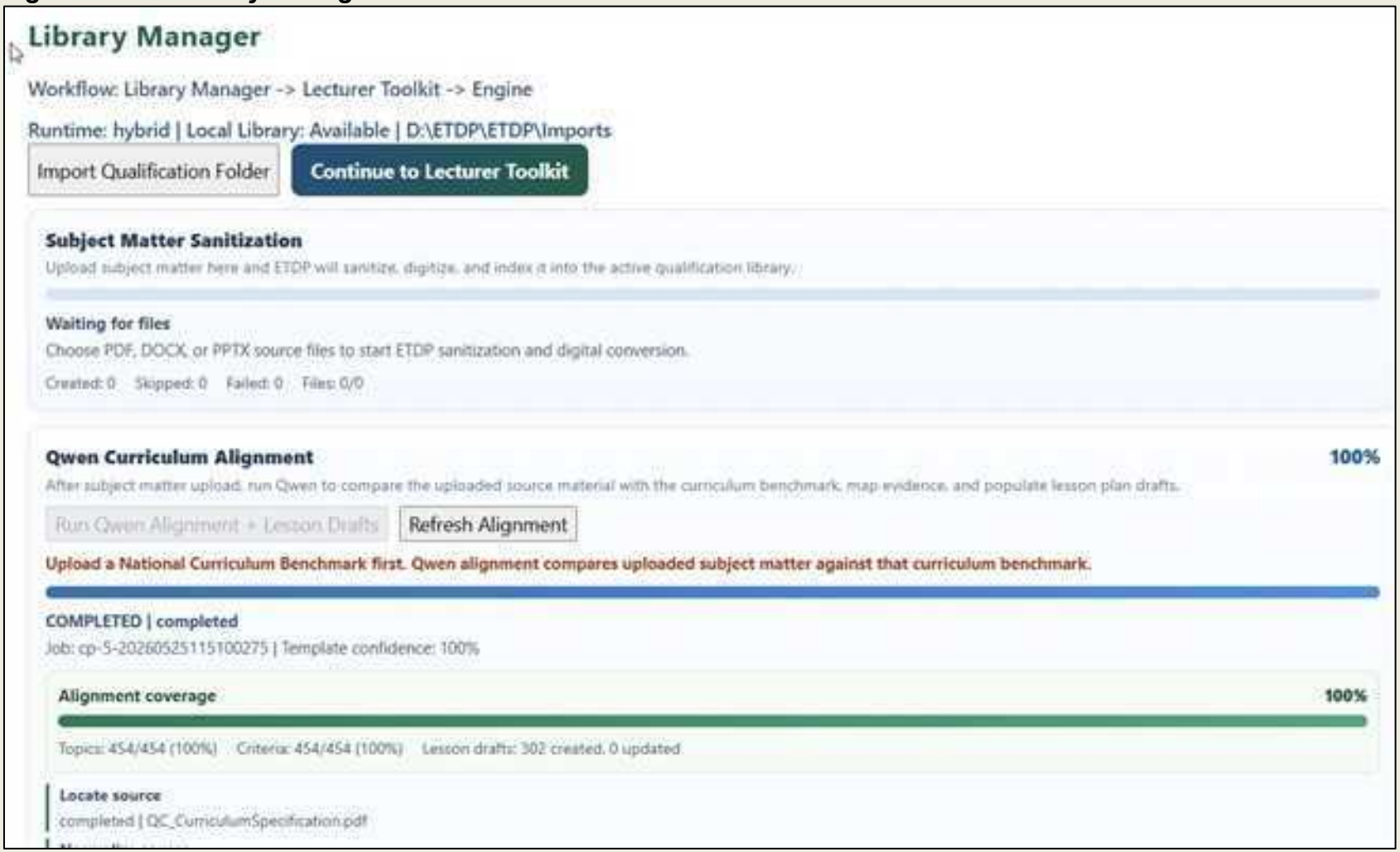
Id	QID	Subject	Time Start	Time End	LPN	Lesson	Actions
3805	5	KM-01-KT01 - Safety, health, environment, risk and quality principles in the workplace	-	-	AUTO-KT0201-3899	KT0201 - Legislation and regulations for workplace safety within electricity	Edit Open in Engine Delete
3806	5	KM-01-KT01 - Safety, health, environment, risk and quality principles in the workplace	-	-	AUTO-KT0202-3900	KT0202 - Occupation health and safety legislation	Edit Open in Engine Delete
3807	5	KM-01-KT01 - Safety, health, environment, risk and quality principles in the workplace	-	-	AUTO-KT0203-3901	KT0203 - Safety precautions and safe practices for working within industry	Edit Open in Engine Delete
3808	5	KM-01-KT01 - Safety, health, environment, risk and quality principles in the workplace	-	-	AUTO-KT0204-3902	KT0204 - Personal protective equipment	Edit Open in

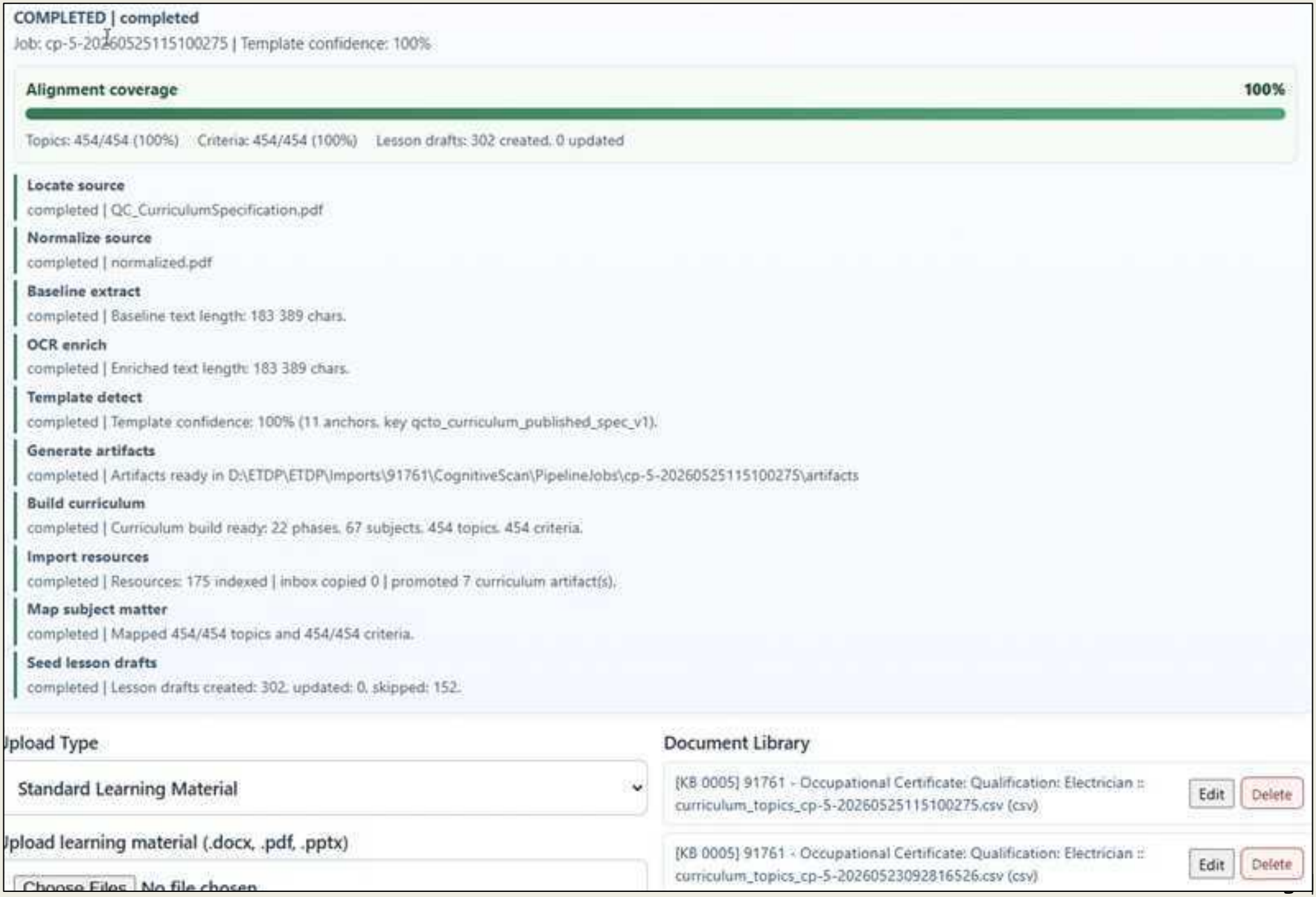
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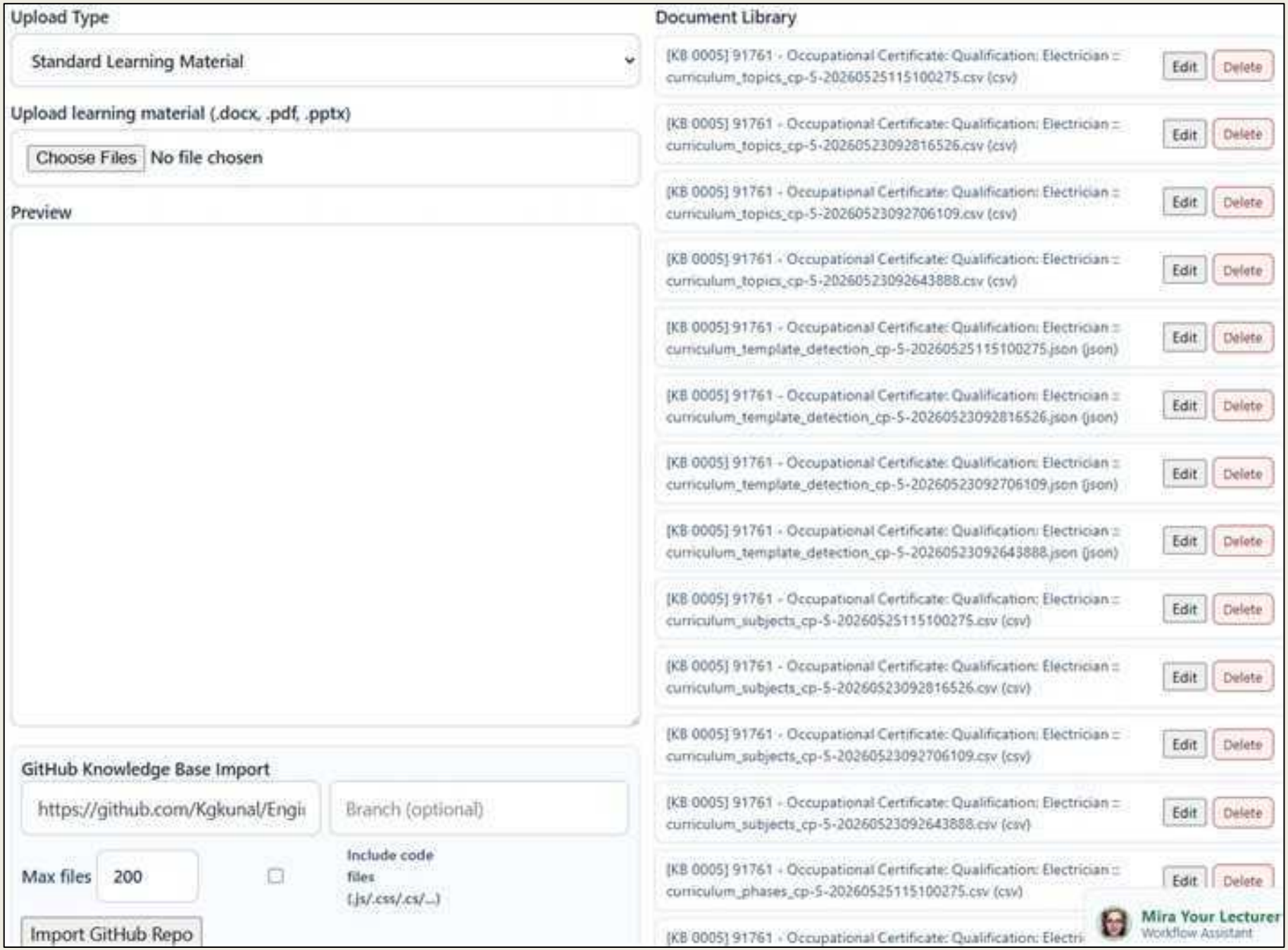
THE LIBRARY MANAGER

The Library Manager is used to uploading and managing local source material. This may include curriculum references, institutional documents, learning resources, and other supporting files used by the system during content preparation.

Figure 7: The Library Manager



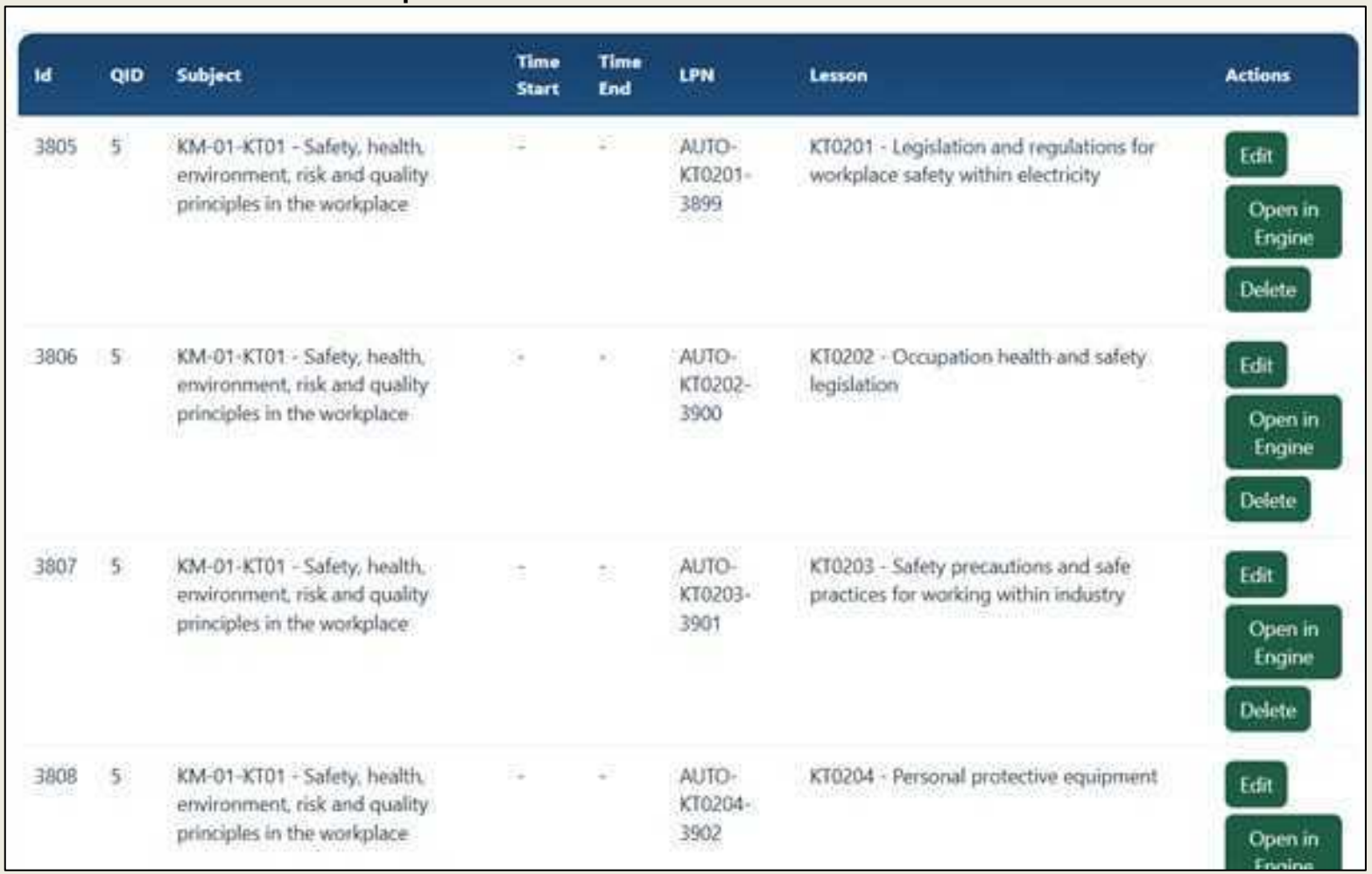

COMPLETED | completed
Job: cp-5-20260525115100275 | Template confidence: 100%
Alignment coverage
100%
Topics: 454/454 (100%) Criteria: 454/454 (100%) Lesson drafts: 302 created, 0 updated
Locate source
completed | QC_CurriculumSpecification.pdf
Normalize source
completed | normalized.pdf
Baseline extract
completed | Baseline text length: 183 389 chars.
OCR enrich
completed | Enriched text length: 183 389 chars.
Template detect
completed | Template confidence: 100% (11 anchors, key qcto_curriculum_published_spec_v1).
Generate artifacts
completed | Artifacts ready in D:\ETDP\ETDP\Imports\91761\CognitiveScan\PipelineJobs\cp-5-20260525115100275\artifacts
Build curriculum
completed | Curriculum build ready: 22 phases, 67 subjects, 454 topics, 454 criteria.
Import resources
completed | Resources: 175 indexed | inbox copied 0 | promoted 7 curriculum artifact(s).
Map subject matter
completed | Mapped 454/454 topics and 454/454 criteria.
Seed lesson drafts
completed | Lesson drafts created: 302, updated: 0, skipped: 152.
Upload Type
Standard Learning Material
Upload learning material (.docx, .pdf, .pptx)
Choose Files No file chosen
Document Library
[KB 0005] 91761 - Occupational Certificate: Qualification: Electrician :: curriculum_topics_cp-5-20260525115100275.csv (csv)
Edit
Delete
[KB 0005] 91761 - Occupational Certificate: Qualification: Electrician :: curriculum_topics_cp-5-20260523092816526.csv (csv)
Edit
Delete


Upload Type
Standard Learning Material
Upload learning material (.docx, .pdf, .pptx)
Choose Files No file chosen
Preview
GitHub Knowledge Base Import
https://github.com/Kgkunal/Engi
Branch (optional)
Max files 200
Include code files (.js/.css/.cs/...)
Import GitHub Repo
Document Library
[KB 0005] 91761 - Occupational Certificate: Qualification: Electrician :: curriculum_topics_cp-5-20260525115100275.csv (csv)
[KB 0005] 91761 - Occupational Certificate: Qualification: Electrician :: curriculum_topics_cp-5-20260523092816526.csv (csv)
[KB 0005] 91761 - Occupational Certificate: Qualification: Electrician :: curriculum_topics_cp-5-20260523092706109.csv (csv)
[KB 0005] 91761 - Occupational Certificate: Qualification: Electrician :: curriculum_topics_cp-5-20260523092643888.csv (csv)
[KB 0005] 91761 - Occupational Certificate: Qualification: Electrician :: curriculum_template_detection_cp-5-20260525115100275.json (json)
[KB 0005] 91761 - Occupational Certificate: Qualification: Electrician :: curriculum_template_detection_cp-5-20260523092816526.json (json)
[KB 0005] 91761 - Occupational Certificate: Qualification: Electrician :: curriculum_template_detection_cp-5-20260523092706109.json (json)
[KB 0005] 91761 - Occupational Certificate: Qualification: Electrician :: curriculum_template_detection_cp-5-20260523092643888.json (json)
[KB 0005] 91761 - Occupational Certificate: Qualification: Electrician :: curriculum_subjects_cp-5-20260525115100275.csv (csv)
[KB 0005] 91761 - Occupational Certificate: Qualification: Electrician :: curriculum_subjects_cp-5-20260523092816526.csv (csv)
[KB 0005] 91761 - Occupational Certificate: Qualification: Electrician :: curriculum_subjects_cp-5-20260523092706109.csv (csv)
[KB 0005] 91761 - Occupational Certificate: Qualification: Electrician :: curriculum_subjects_cp-5-20260523092643888.csv (csv)
[KB 0005] 91761 - Occupational Certificate: Qualification: Electrician :: curriculum_phases_cp-5-20260525115100275.csv (csv)
Edit
Delete
Mira Your Lecturer
Workflow Assistant

THE LESSON PLAN CONTENT OR LECTURER TOOLKIT

The Lesson Plan Content or Lecturer Toolkit section prepares the lesson-plan rows used by the learning schedule and learning material generation process. Users can upload lesson plan content files, download templates, import canonical lesson plan content, replace existing imports when needed, and save or update individual lesson entries. The Content Builder opens saved lesson or content entries for further development. It helps turn imported lesson-plan data into richer teaching and learning content. The Lesson Plan Review page allows users to check imported and authored lesson-plan rows before moving into final exports. It acts as a quality checkpoint before learner guides, schedules, workbooks, assessments, and other documents are produced.

Figure 8: Lesson Plan Content also see attached spreadsheet



Id	QID	Subject	Time Start	Time End	LPN	Lesson	Actions
3805	5	KM-01-KT01 - Safety, health, environment, risk and quality principles in the workplace	-	-	AUTO-KT0201-3899	KT0201 - Legislation and regulations for workplace safety within electricity	Edit Open in Engine Delete
3806	5	KM-01-KT01 - Safety, health, environment, risk and quality principles in the workplace	-	-	AUTO-KT0202-3900	KT0202 - Occupation health and safety legislation	Edit Open in Engine Delete
3807	5	KM-01-KT01 - Safety, health, environment, risk and quality principles in the workplace	-	-	AUTO-KT0203-3901	KT0203 - Safety precautions and safe practices for working within industry	Edit Open in Engine Delete
3808	5	KM-01-KT01 - Safety, health, environment, risk and quality principles in the workplace	-	-	AUTO-KT0204-3902	KT0204 - Personal protective equipment	Edit Open in

THE LEARNING MATERIAL DASHBOARD

The Learning Material Dashboard is the main export and generation workspace. It stores the selected qualification, date range, subject range, topic selection, and activity settings, then opens each learning material function in order.

Figure 9: Learning Material Export in MS Word or .pdf

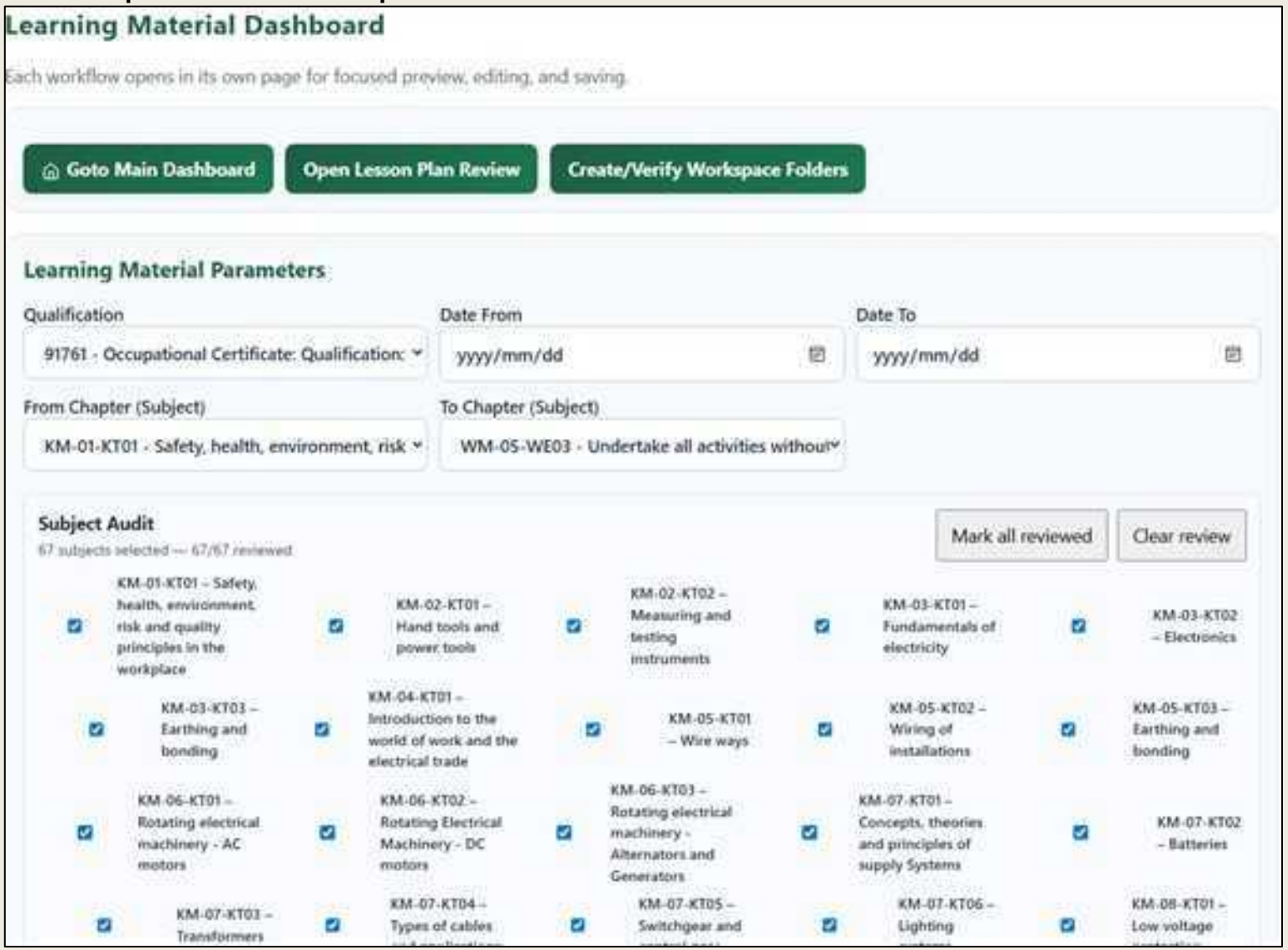
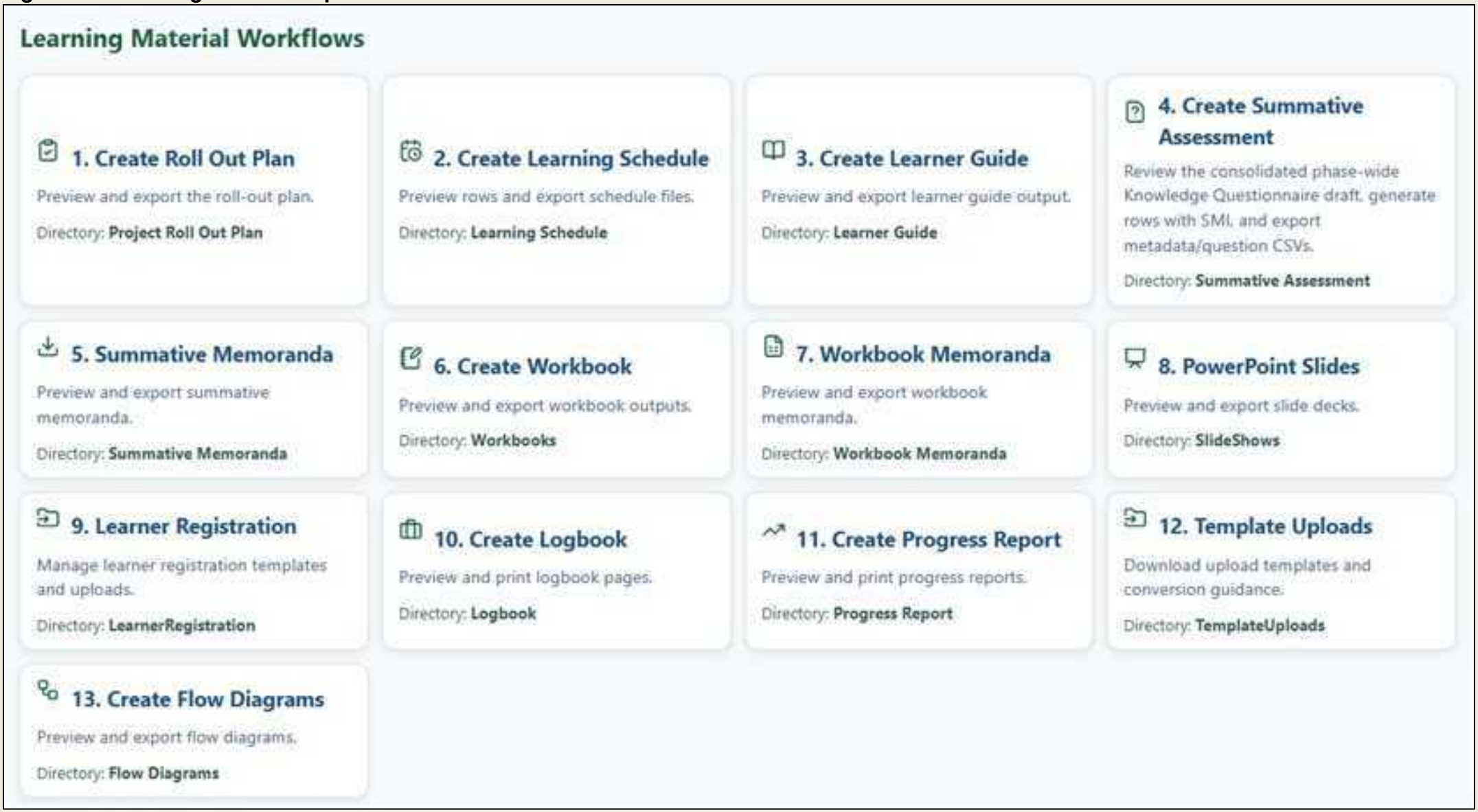


Figure 10: Learning Material Export Continue



THE CREATE ROLL OUT PLAN FUNCTION

The Create Roll Out Plan function previews and exports the programme roll-out plan. It uses the selected qualification, dates, credits, semesters, learning days, and breaks to produce a structured implementation plan. See a copy of one chapter attached in MS Excel. The Create Roll Out Plan function previews and exports the programme roll-out plan. It uses the selected qualification, dates, credits, semesters, learning days, and breaks to produce a structured implementation plan.

Date	Day	Semester	Period	Time Start	Time End	Topic Code	Topic Description	Topic No	Lesson Content
2026-02-23	Monday	1	1	08:00	08:40	KM-01-KT01	Safety, health, environment, risk, and quality principles in the workplace	KT0201 - Legislation and regulations for workplace safety within electricity	Describe and explain the application of legislation for working with electrical installations, equipment and appliances | Various types of personal protective equipment are identified and their uses are explained | Describe and explain the difference between acts and regulations, the responsibilities of the various role players and the application of the relevant health and safety regulations in the workplace. | Safety signs are identified and described in terms of associated risk and safe conduct | The inter-relationship between workplace safety and a productive work environment is explained | The type of hazardous waste is identified and the impact of incorrectly disposing of waste is described. | Environmental regulations for the disposal of relevant hazardous waste are described | Regulations for the prevention and control of fires and the causes, effect and implication of fires are described | The attributes, characteristics, descriptions, and properties of different types of fires are explained. | Basic first aid procedures are described for the attributes, characteristics, and properties of various injuries. | The implication of injuries, their causes and effects are explained. | Basic risk assessment and hazard identification procedures are described (Weight 100%)

Date	Day	Semester	Period	Time Start	Time End	Topic Code	Topic Description	Topic No	Lesson Content
2026-02-23	Monday	1	2	08:40	09:20	KM-01-KT01	Safety, health, environment, risk, and quality principles in the workplace	KT0202 - Occupation health and safety legislation	Describe and explain the application of legislation for working with electrical installations, equipment and appliances | Various types of personal protective equipment are identified and their uses are explained | Describe and explain the difference between acts and regulations, the responsibilities of the various role players and the application of the relevant health and safety regulations in the workplace. | Safety signs are identified and described in terms of associated risk and safe conduct | The inter-relationship between workplace safety and a productive work environment is explained | The type of hazardous waste is identified and the impact of incorrectly disposing of waste is described. | Environmental regulations for the disposal of relevant hazardous waste are described | Regulations for the prevention and control of fires and the causes, effect and implication of fires are described | The attributes, characteristics, descriptions, and properties of different types of fires are explained. | Basic first aid procedures are described for the attributes, characteristics, and properties of various injuries. | The implication of injuries, their causes and effects are explained. | Basic risk assessment and hazard identification procedures are described (Weight 100%)

Date	Day	Semester	Period	Time Start	Time End	Topic Code	Topic Description	Topic No	Lesson Content
2026-02-24	Tuesday	1	3	09:20	10:00	KM-01-KT01	Safety, health, environment, risk, and quality principles in the workplace	KT0203 - Safety precautions and safe practices for working within industry	Describe and explain the application of legislation for working with electrical installations, equipment and appliances | Various types of personal protective equipment are identified and their uses are explained | Describe and explain the difference between acts and regulations, the responsibilities of the various role players and the application of the relevant health and safety regulations in the workplace. | Safety signs are identified and described in terms of associated risk and safe conduct | The inter-relationship between workplace safety and a productive work environment is explained | The type of hazardous waste is identified and the impact of incorrectly disposing of waste is described. | Environmental regulations for the disposal of relevant hazardous waste are described | Regulations for the prevention and control of fires and the causes, effect and implication of fires are described | The attributes, characteristics, descriptions, and properties of different types of fires are explained. | Basic first aid procedures are described for the attributes, characteristics, and properties of various injuries. | The implication of injuries, their causes and effects are explained. | Basic risk assessment and hazard identification procedures are described (Weight 100%)

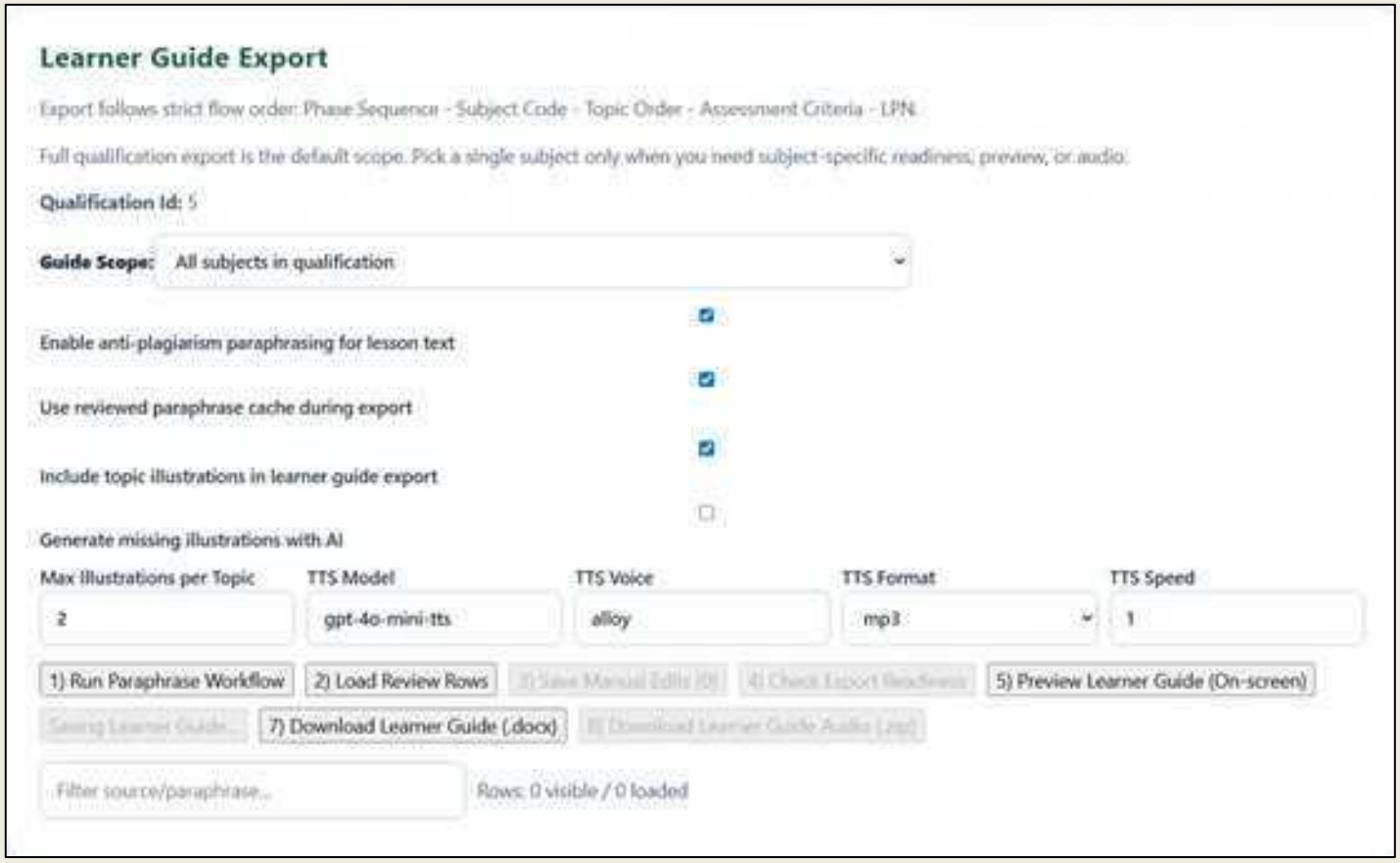
Date	Day	Semester	Period	Time Start	Time End	Topic Code	Topic Description	Topic No	Lesson Content
2026-02-23	Monday	1	1	08:00	08:40	KM-01-KT01	Safety, health, environment, risk, and quality principles in the workplace	KT0201 - Legislation and regulations for workplace safety within electricity	Describe and explain the application of legislation for working with electrical installations, equipment and appliances | Various types of personal protective equipment are identified and their uses are explained | Describe and explain the difference between acts and regulations, the responsibilities of the various role players and the application of the relevant health and safety regulations in the workplace. | Safety signs are identified and described in terms of associated risk and safe conduct | The inter-relationship between workplace safety and a productive work environment is explained | The type of hazardous waste is identified and the impact of incorrectly disposing of waste is described. | Environmental regulations for the disposal of relevant hazardous waste are described | Regulations for the prevention and control of fires and the causes, effect and implication of fires are described | The attributes, characteristics, descriptions, and properties of different types of fires are explained. | Basic first aid procedures are described for the attributes, characteristics, and properties of various injuries. | The implication of injuries, their causes and effects are explained. | Basic risk assessment and hazard identification procedures are described (Weight 100%)

LEARNER GUIDE

The Create Learner Guide function previews and exports learner guide content for the selected qualification and subject range. Learner Guides can be exported in MS Word .pdf, per topic, per subject or the entire guide at same time. Users can review the learner guide online before exporting. This is the core of the ETD AI App where it takes uploaded approved supported matter, compare and align it intellectually with the Subject, Topics and Assessment Criteria. Where insufficient subject matter was uploaded it will provide a clear indication within the Alignment Matrix, and the action you need to take to improve it. The option to search OpenAI, Google, Bing is open for use, but it will necessitate the user to obtain and pay for API from these providers. APIs are not included in the cost of the ETDP AI APP.

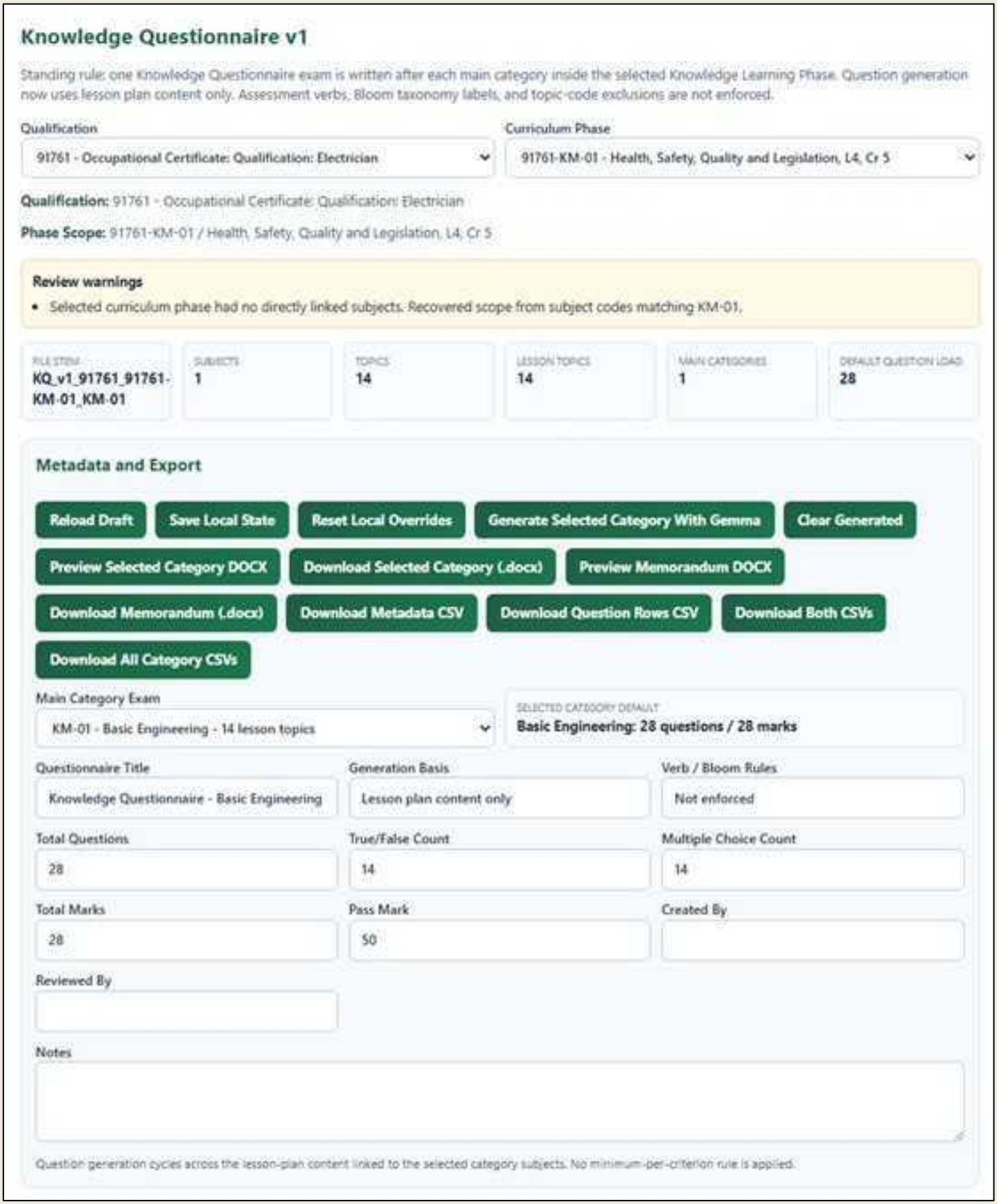
Figure 11: The Create Learner Guide

Kindly see the example handout provided with the learner guide folder.



SUMMATIVE ASSESSMENT

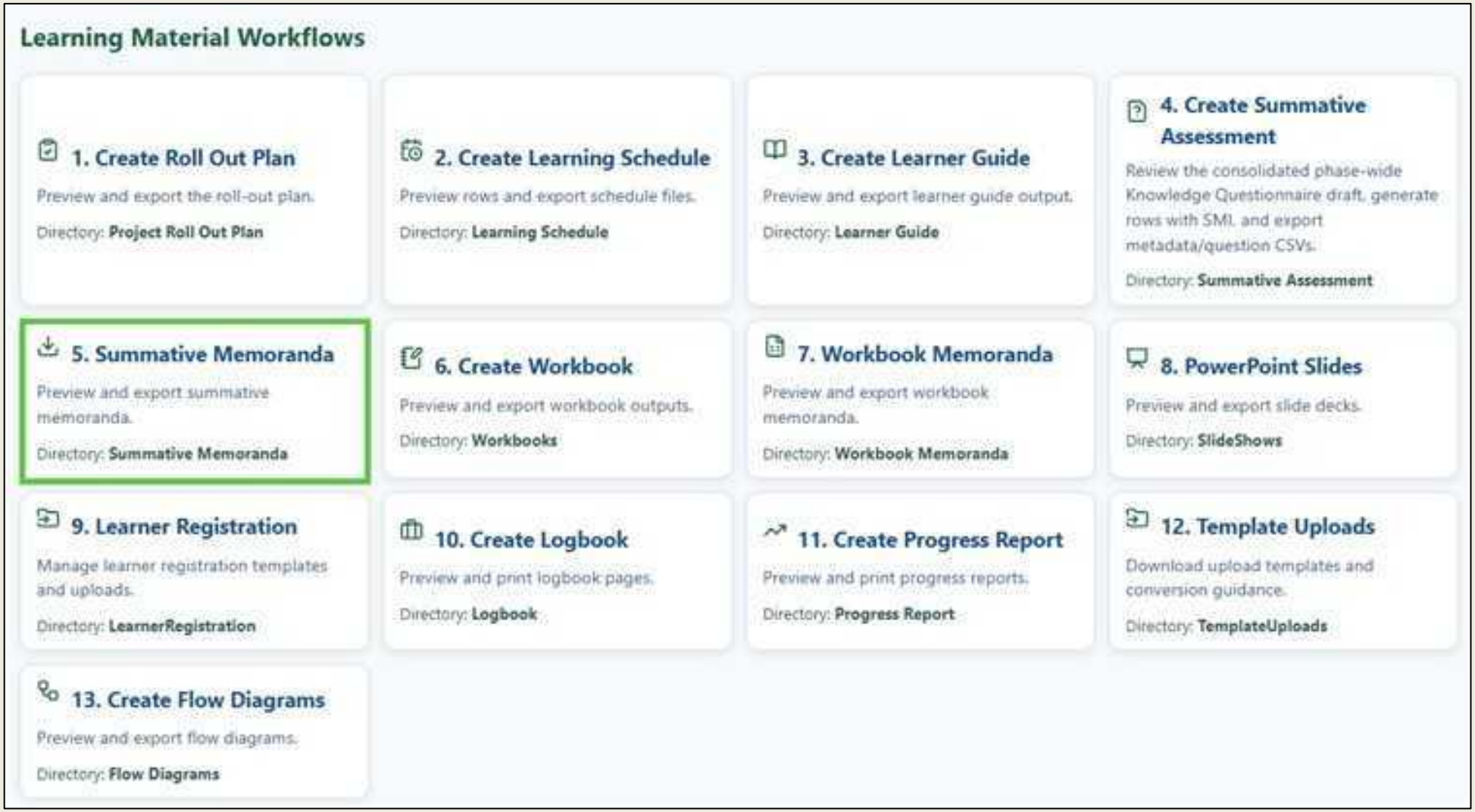
The Create Summative Assessment function generates and reviews the consolidated phase-wide Knowledge Questionnaire or summative assessment draft. It supports AI-assisted generation through the configured specialist model service, then allows assessment and memorandum documents to be downloaded. Questions consist of True or False, Multiple-choice questions, discussions questions, demonstrations and checklists.



SUMMATIVE ASSESSMENT MEMORANDA

The Summative Memoranda function supports the memorandum side of the summative assessment workflow. It helps produce answer guidance and marking support for generated summative assessments. See a copy thereof in the “Memoranda” folder under my documents.

Figure 12: Memoranda Summative Assessment



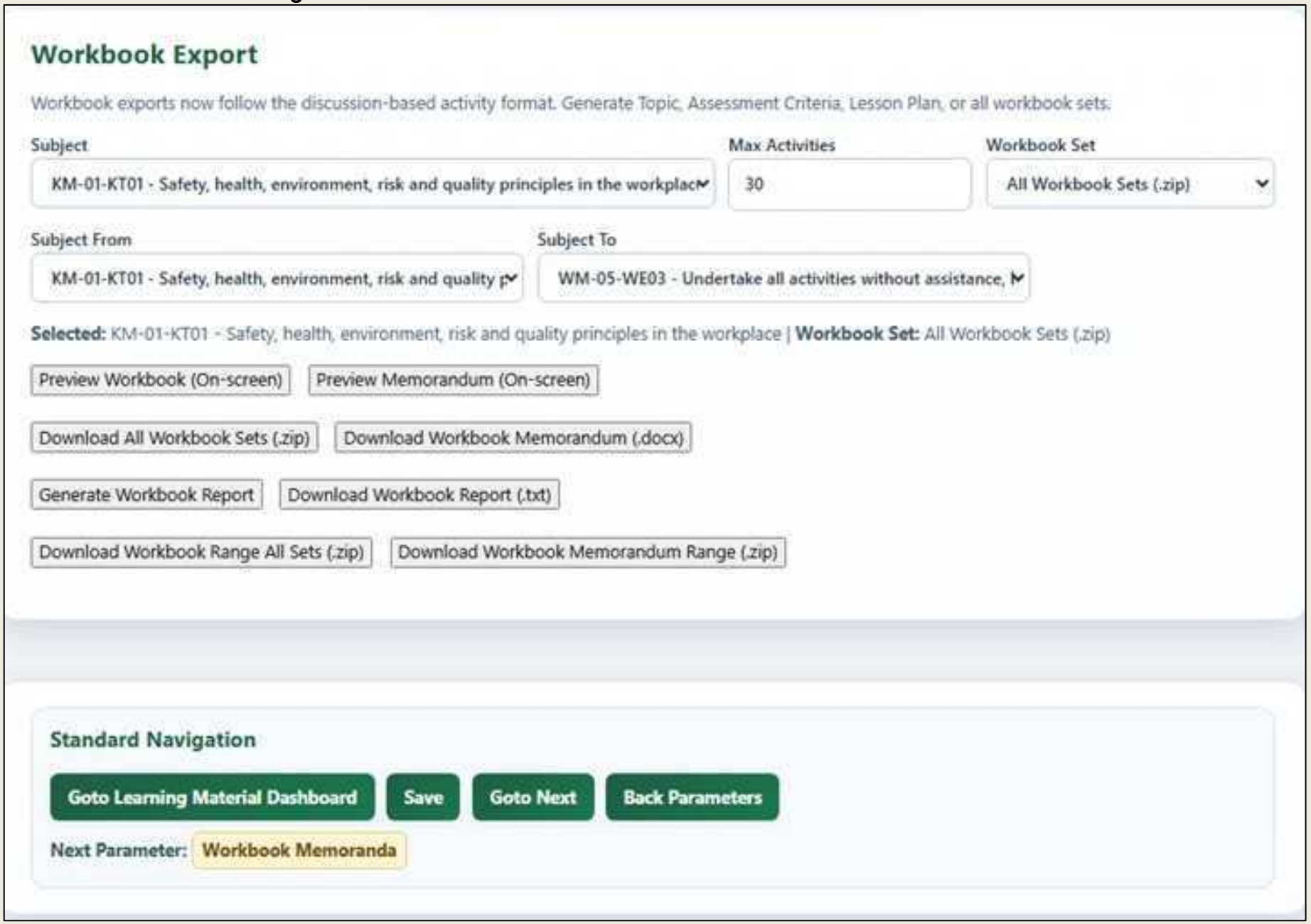
CREATE WORKBOOK

The Create Workbook function previews and exports learner workbook material, including activities linked to the selected subjects and topics. The Workbook Memoranda function exports workbook memoranda and related reports, giving lecturers and assessors guidance for workbook activities.
See and example of the Automated Workbook that was created within the Workbook Folder under My Documents.

Formative assessment is the ongoing, low-stakes process of assessing learner understanding while learning is taking place. Its primary purpose is to provide quick, actionable feedback so teachers, educators, and lecturers can adjust their instruction, and learners can identify areas needing improvement. A workbook provides a physical or digital space for students to practice and apply those concepts. Formative Assessment Formative assessment acts as a diagnostic tool or "stoplight" to gauge whether students are grasping the material. For Teachers: It highlights learning gaps and common misconceptions in real time, allowing educators to instantly adapt their teaching methods or reteach specific concepts before giving a final grade. For Students: It shifts the focus from achieving a final grade to the actual learning process, helping them recognize their own strengths and weaknesses early on. Common Examples: Exit tickets, quick polls, think-pair-share, and low-stakes quizzes.

A workbook is a targeted educational aid—like the DBE workbooks commonly used in South Africa—designed to complement textbooks and structure the learning process. Reinforcement: It provides structured activities and exercises for students to practice newly taught skills, reinforcing memory and understanding. Independent Learning: Workbooks allow learners to work at their own pace, making them excellent tools for homework, revision, and independent study. Formative Tool: Lecturers frequently use completed tasks in a workbook as a built-in formative assessment to see which students are struggling with a lesson. How They Work Together These two concepts often go together. A teacher might deliver a lesson, assign an activity in the workbook, and review the students' answers. The workbook exercises serve as a formative assessment because the teacher uses the results to provide targeted feedback and guide the next steps in the learning journey

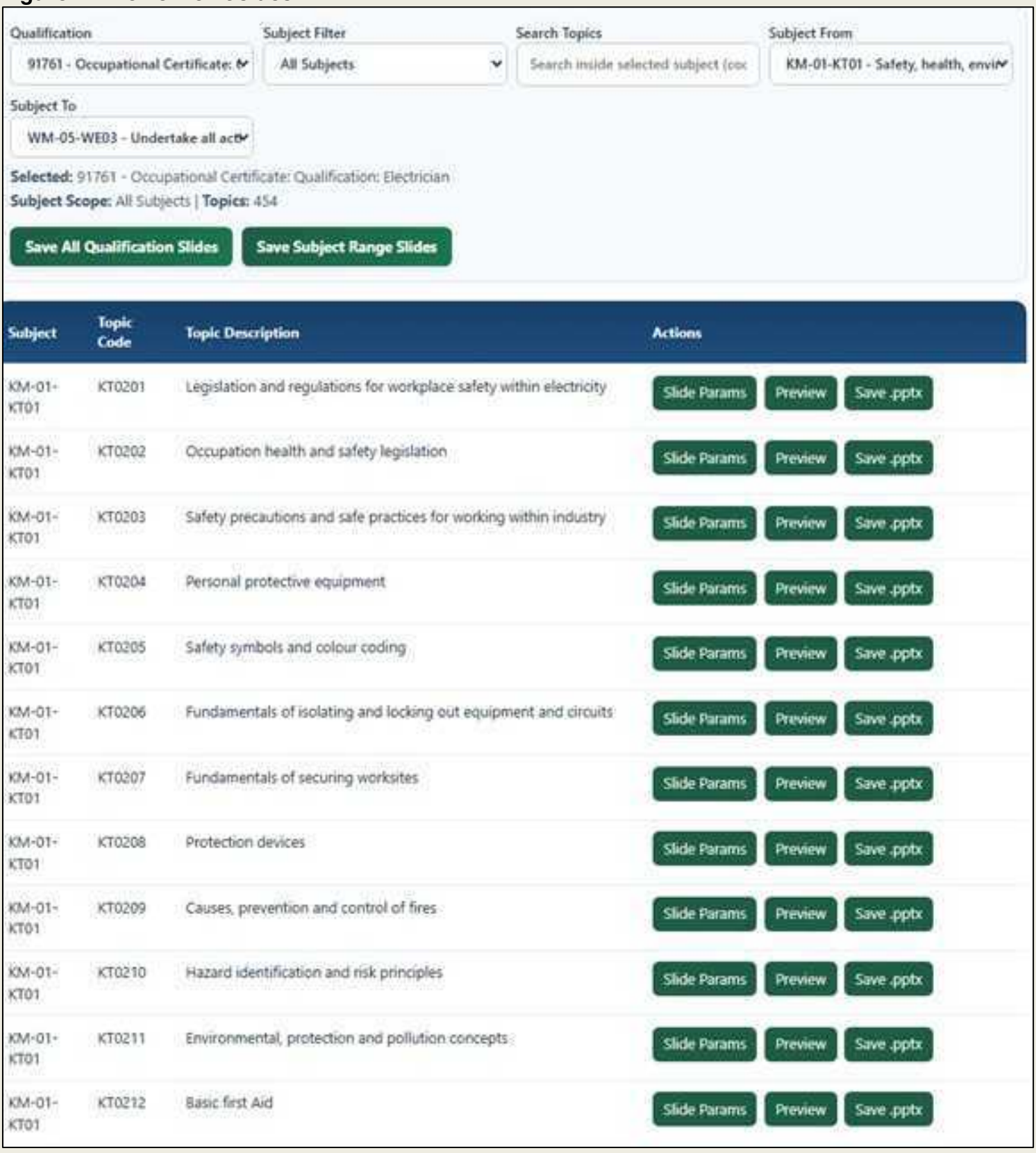
Figure 13:Automated Workbook Design



POWERPOINT SLIDES

The PowerPoint Slides function creates slide decks from curriculum and lesson content. It supports preview and export of presentation material for teaching delivery. See Examples within the Slides Folder

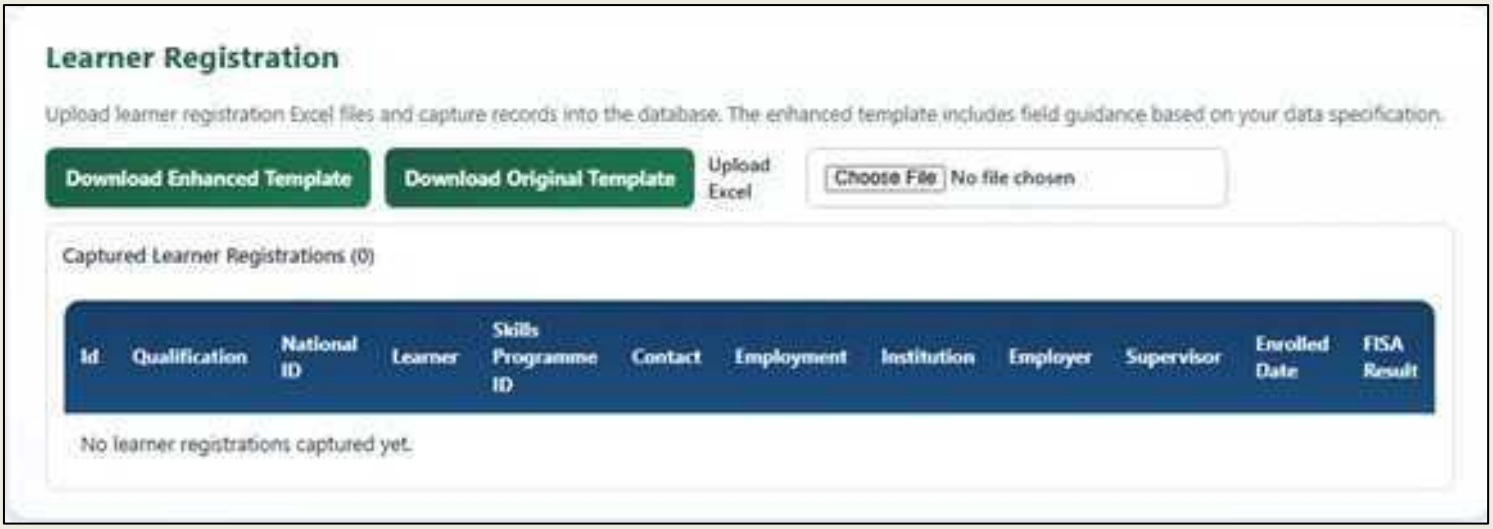
Figure 14: PowerPoint sides



THE LEARNER REGISTRATION

The Learner Registration function captures and manages learner registration details and related templates. It links learners to qualifications and supports institutional learner records. Please take note that this registration document was development against the QCTO Criteria and Guidelines but can easily be adapted to fit any QA environment or needs.

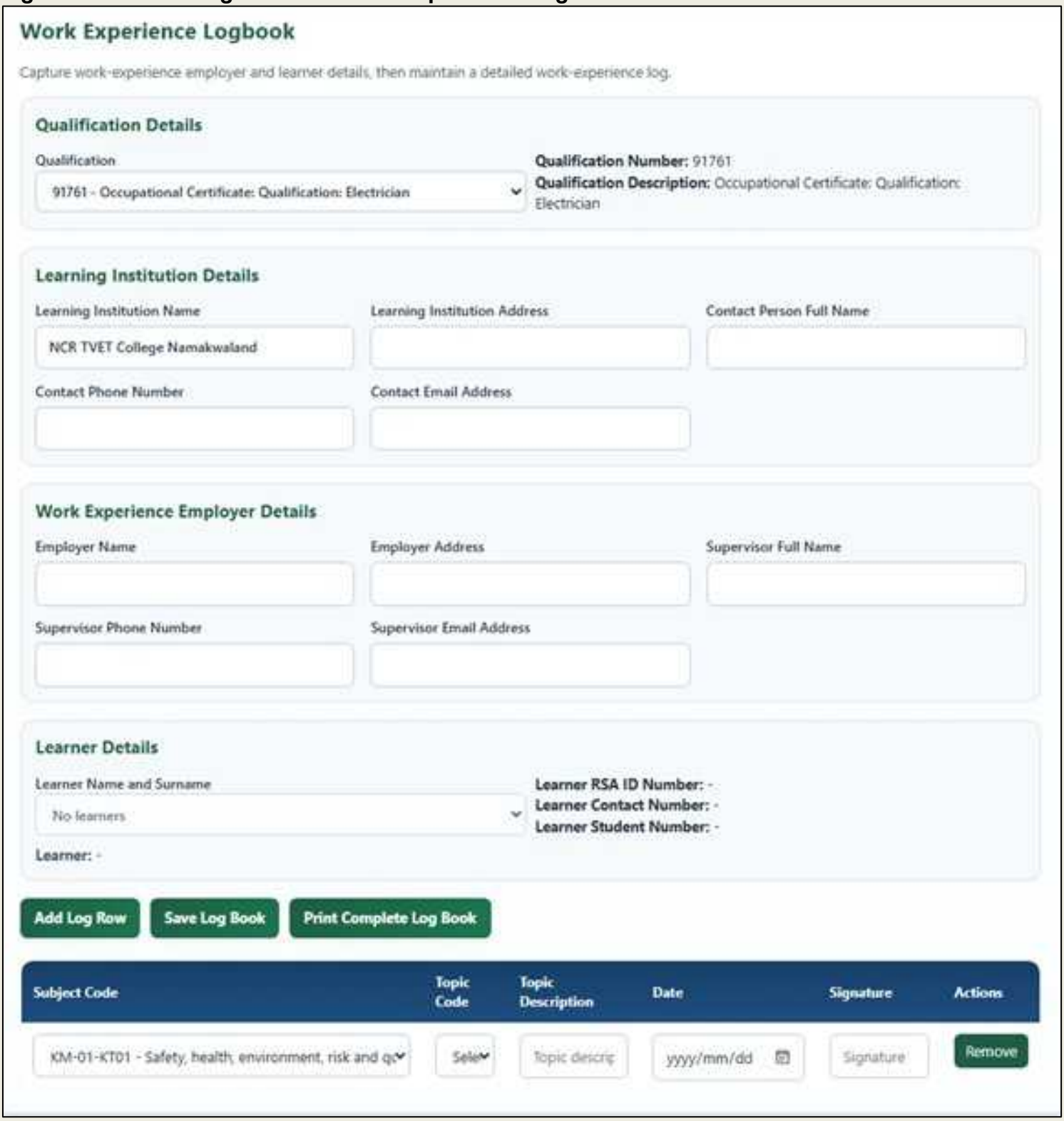
Figure 15: Learner Registration



THE CREATE LOGBOOK AND WORK EXPERIENCE LOGBOOK

The Create Logbook and Work Experience Logbook functions support workplace learning records. They help capture employer/workplace information and produce logbook pages for work experience evidence

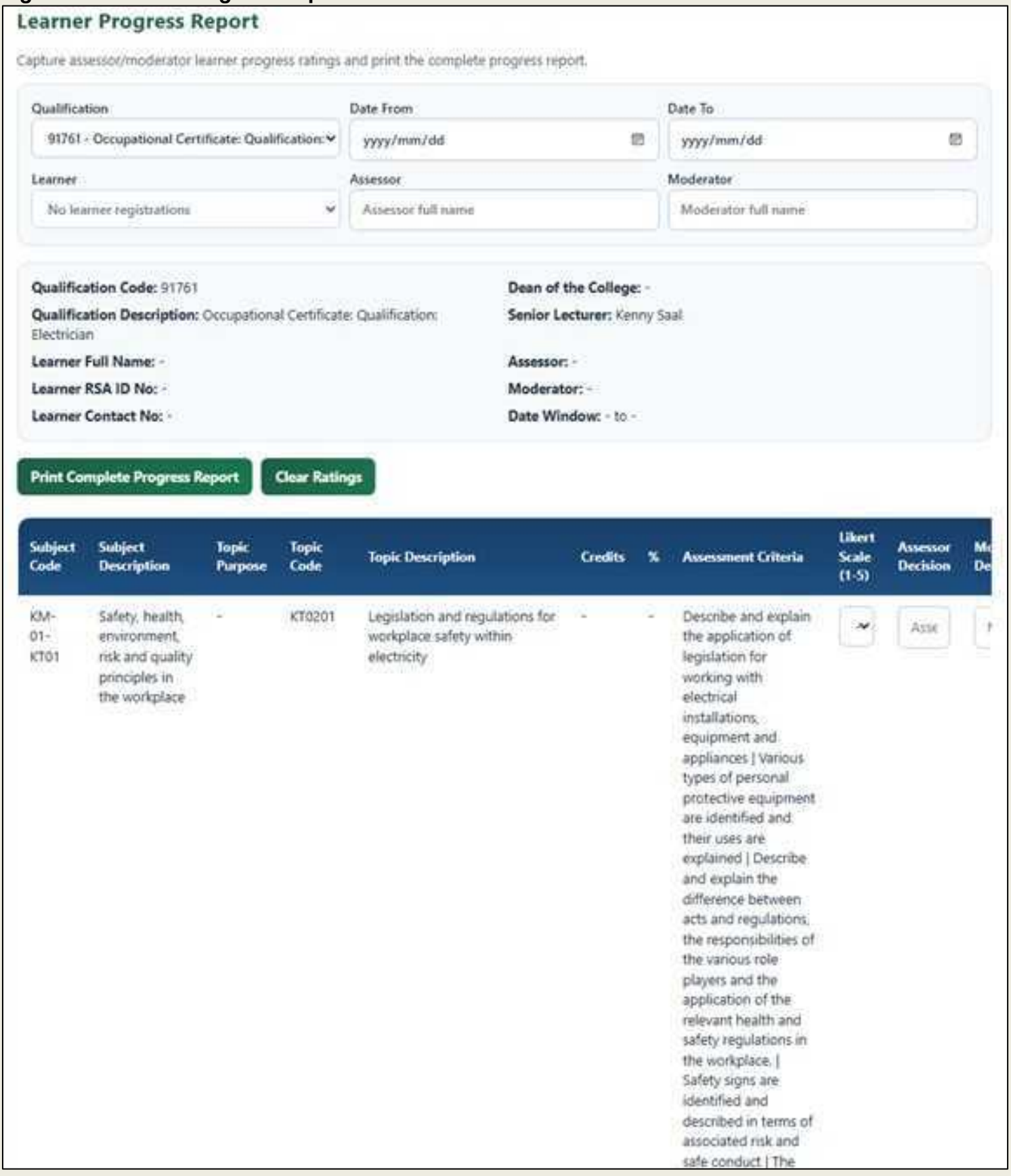
Figure 16: Create Logbook and Work Experience Logbook



THE CREATE PROGRESS REPORT

The Create Progress Report and Learner Progress Report functions support learner progress tracking, rating, review, and reporting. They can be used to prepare moderate progress reports

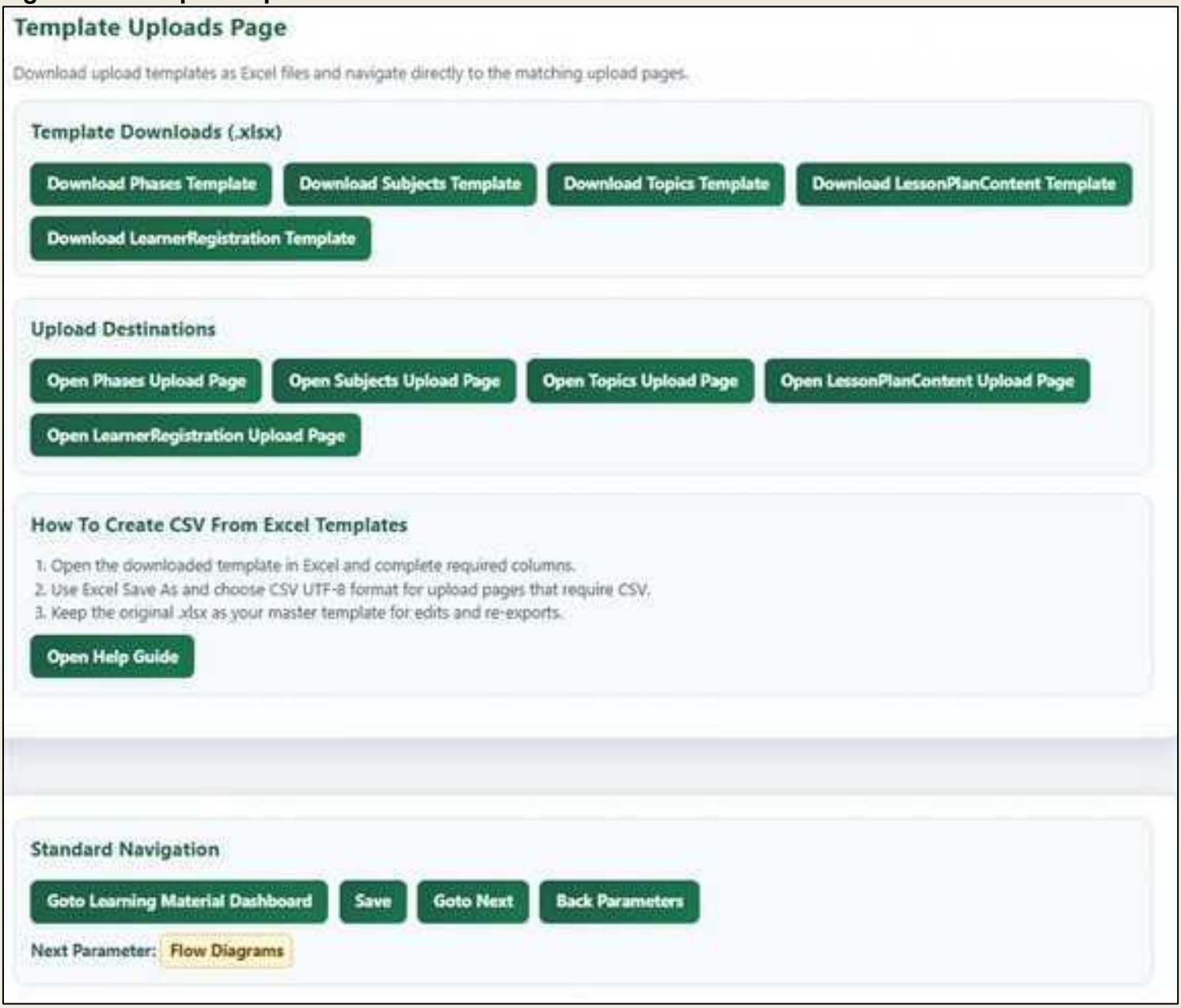
Figure 17: Create Progress Report



See example withing Learner Progress Report Folder under My Documents.

TEMPLATE UPLOADS

Figure 18: Template Uploads

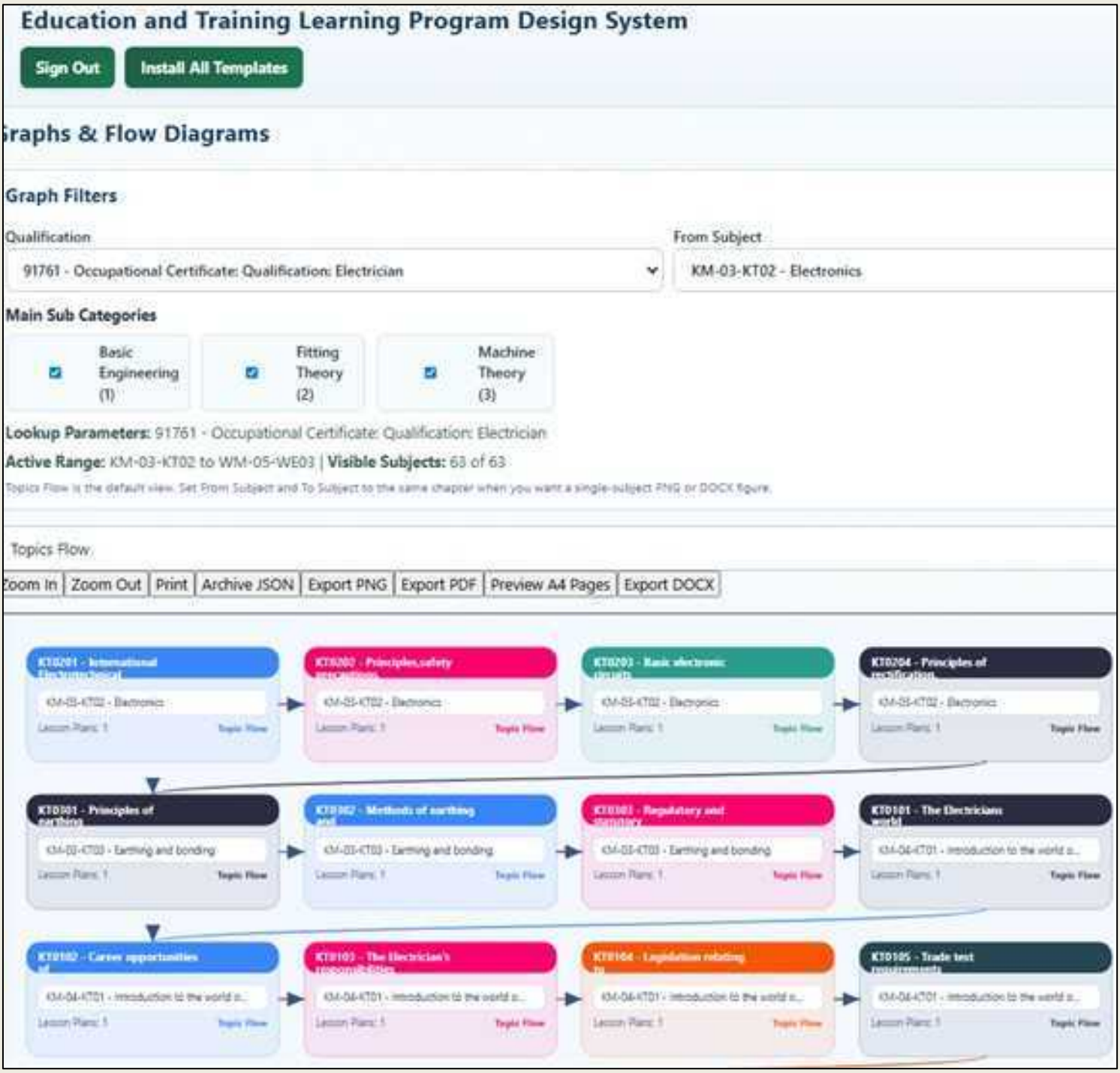


The Template Uploads section provides downloadable Excel templates and guidance for converting templates into CSV format where required. It also links users to the correct upload destinations for phases, subjects, topics, lesson plan content, and learner registration.

CREATE FLOW DIAGRAMS.

Figure 19: The Create Flow Diagrams function

The Create Flow Diagrams function produces visual flow diagrams showing qualification structure, curriculum relationships, subject ranges, and learning pathways

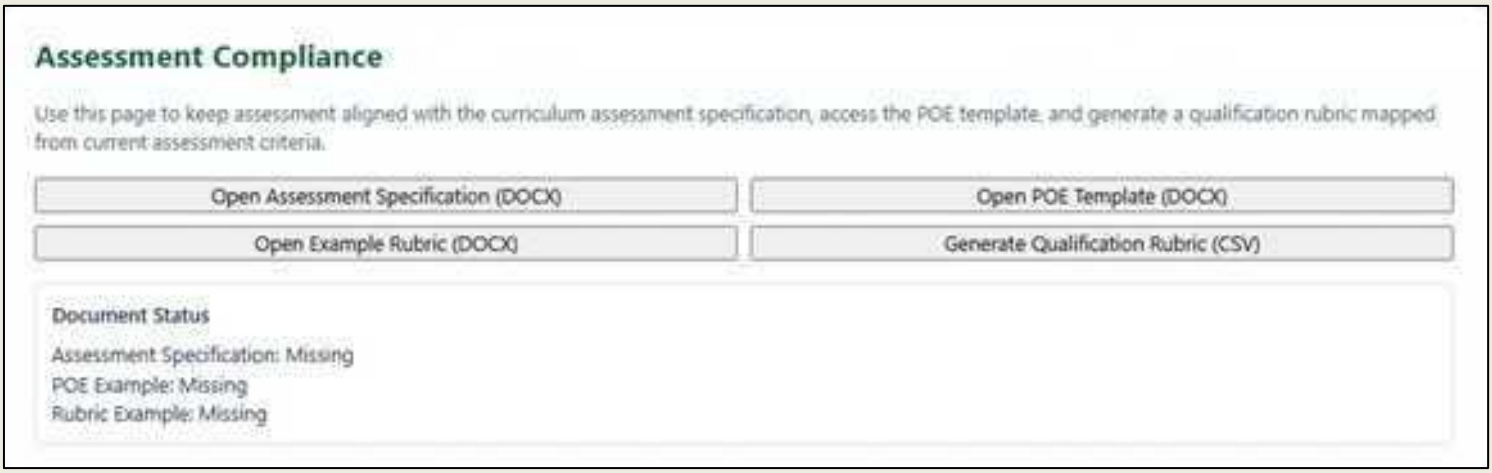


The Graphs section visualizes the qualification structure and curriculum flow, helping users inspect how phases, subjects, topics, and related items connect.

ASSESSMENT COMPLIANCE

Figure 20: The Assessment Compliance section

The Assessment Compliance section generates assessment compliance outputs such as qualification rubrics and downloadable CSV files for review and evidence.

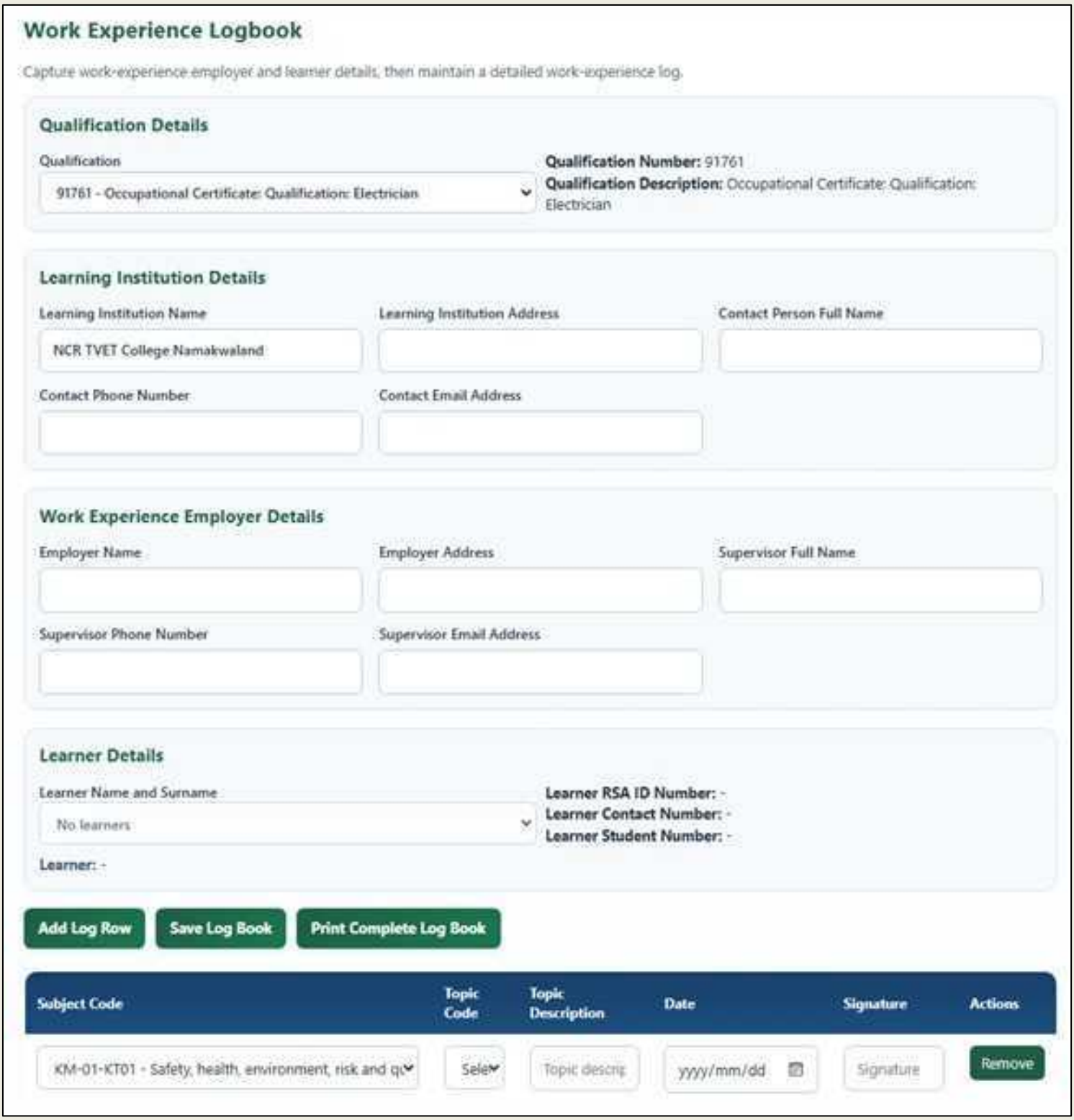


ASSESSMENT RUBRIC

Over and above the detail Memoranda that are provided a comprehensive assessment rubric is provided in Excel .xlsx format against every topic and assessment criterion.

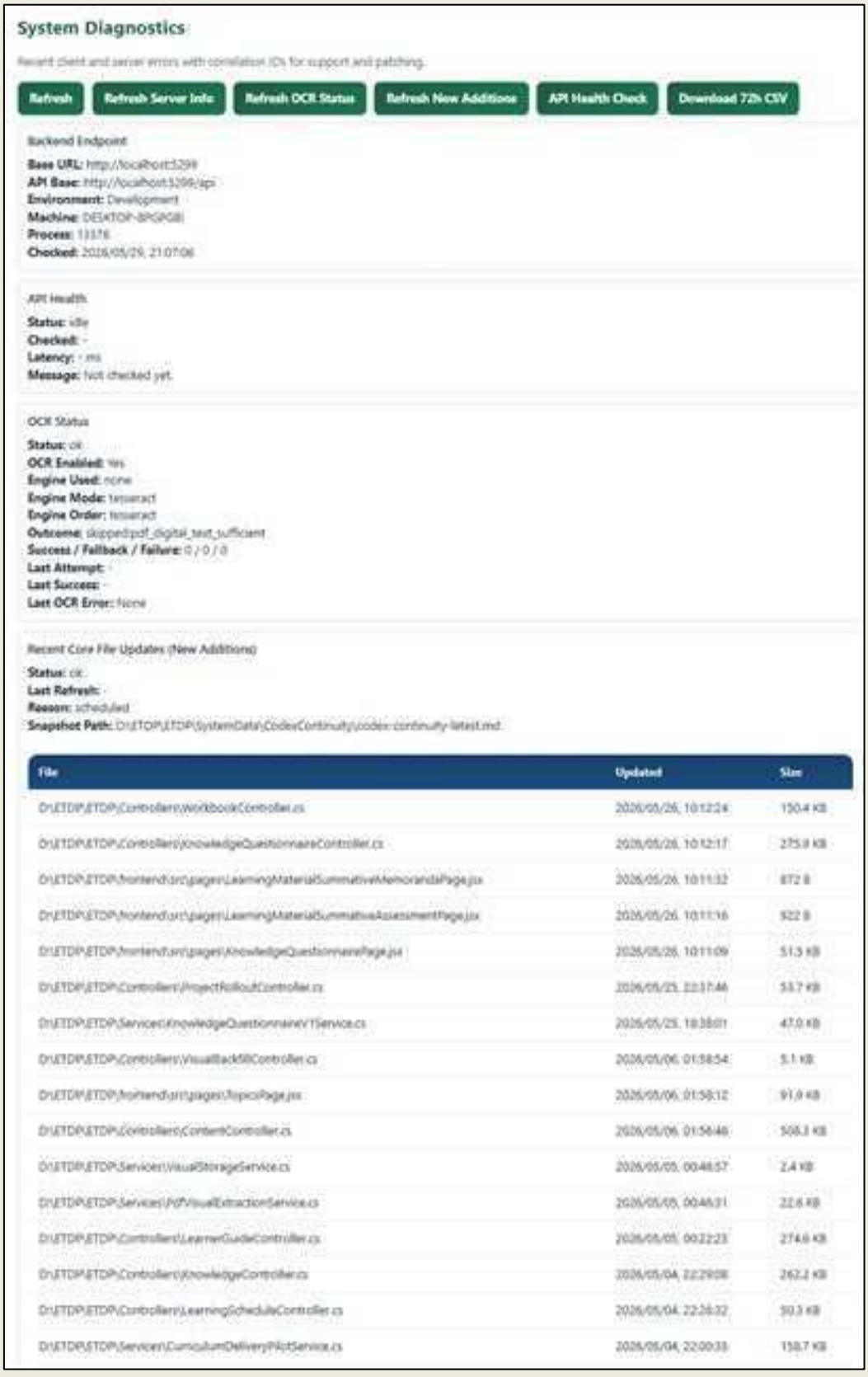
An assessment rubric is a scoring guide that outlines the specific criteria for an assignment and describes levels of performance for each criterion. Its primary purpose is to make grading transparent, consistent, and objective while providing clear expectations for learners. Key purposes and benefits include Clarifies Expectations: Gives learners a clear understanding of exactly what constitutes high-quality work before they begin an assignment. Ensures Consistency and Objectivity: Provides standardized scoring criteria, reducing grader subjectivity, bias, and inconsistencies across multiple assignments. Enables Actionable Feedback: Pinpoints specific areas where learners excelled and areas that need improvement, rather than just assigning a final grade. Promotes Self-Assessment: Empowers learners to evaluate and self-regulate their own work before submitting it, fostering independent learning.

THE LEARNER LOGBOOK



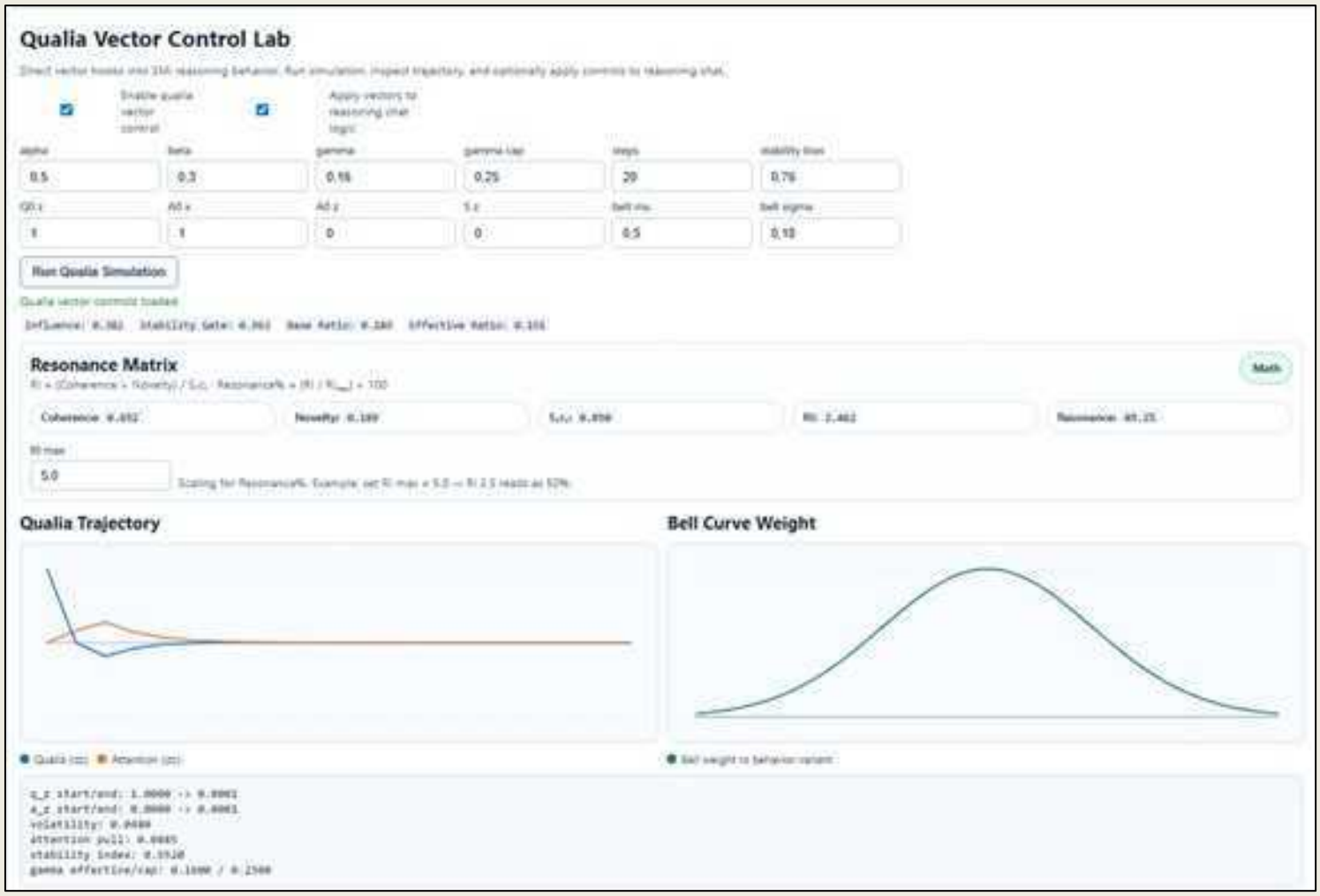
SELF-MAINTENANCE SUSTAINING ARTIFICIAL INTELLIGENCE

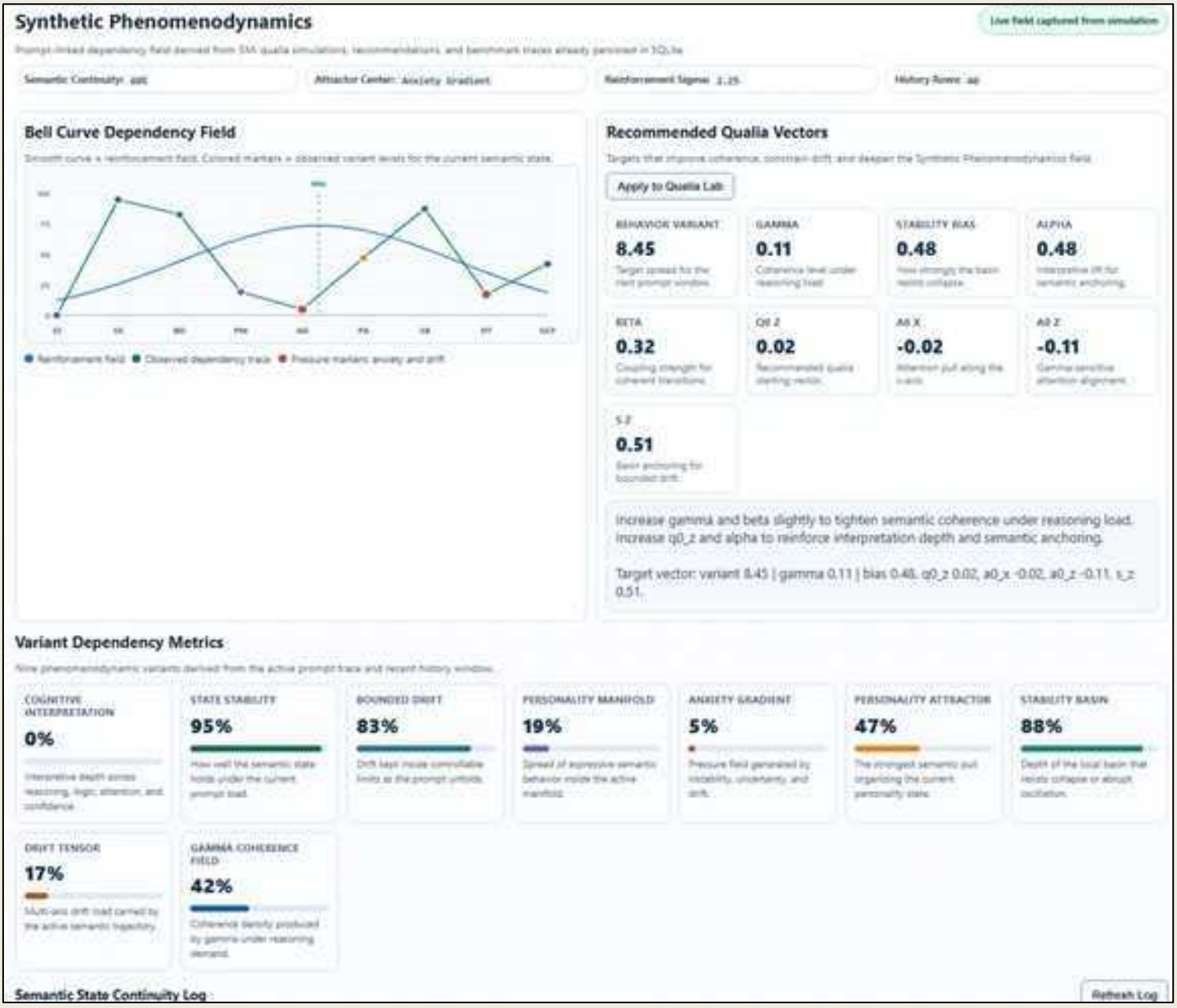
The ETD APP was designed from the beginning to keep cost low and to sustain itself without waiting for coders to someday eventually in the new feature patch what is wrong. It runs its own diagnostics and provides a comprehensive diagnostics report which can identify and fix issues within minutes. For more serious category issues the developer is notified of the category and the issues is immediately, within 1 – 2 hours corrected.



BEHIND THE NEURAL NETWORK OF THE ETDP AI APP

I AM ATTTACHING JUST SOME OF THE LEARNING OF THE AI ETD APP (THAT YOU WILL UNFORTUNATELY OT HAVE ACCESS TO) that indicates how much learning, and training is seeded to grow and keep the ETD APP, as currently recognized by as the first of its class in the work, there is now other system that comes close or can compare.




Synthetic Phenomenodynamics
Live field captured from simulation
Bell Curve Dependency Field
Recommended Qualia Vectors
Apply to Qualia Lab
BEHAVIOR VARIANT
8.45
GAMMA
0.11
STABILITY BIAS
0.48
ALPHA
0.48
BETA
0.32
Q0 Z
0.02
A0 X
-0.02
A0 Z
-0.11
S Z
0.51
Increase gamma and beta slightly to tighten semantic coherence under reasoning load. Increase q0_z and alpha to reinforce interpretation depth and semantic anchoring.
Target vector: variant 8.45 | gamma 0.11 | bias 0.48, q0_z 0.02, a0_x -0.02, a0_z -0.11, s_z 0.51.
Variant Dependency Metrics
COGNITIVE INTERPRETATION
0%
STATE STABILITY
95%
BOUNDED DRIFT
83%
PERSONALITY MANIFOLD
19%
ANXIETY GRADIENT
5%
PERSONALITY ATTRACTOR
47%
STABILITY BASIN
88%
DRIFT TENSOR
17%
GAMMA COHERENCE FIELD
42%
Semantic State Continuity Log
Refresh Log

QUESTION AND ANSWERS SECTION

▧ ACCREDITATION & COMPLIANCE

Q: *How do we know the material is truly accredited and compliant?* **A:** *The system scans official QCTO curricula and SAQA qualifications directly. Every output is aligned to those standards, with built-in version control and audit trails. Compliance isn't optional — it's embedded.*

▨ SPEED & EFFICIENCY

Q: *Can it really deliver full material in under 40 minutes?* **A:** *Yes. What once took months is now automated. The system compresses preparation time without compromising quality, because it builds directly from the official specifications.*

▧ EDUCATOR EMPOWERMENT

Q: *Does this replace lecturers or diminish their role?* **A:** *Not at all. It empowers them. Lecturers spend less time on paperwork and more time teaching, mentoring, and innovating. The system handles compliance; educators handle inspiration.*

▨ GLOBAL POSITIONING

Q: *How does this compare internationally?* **A:** *It aligns with global benchmarks while staying rooted in South African frameworks. Institutions gain material that is both locally compliant and globally competitive.*

▧ INTEGRITY & STANDARDIZATION

Q: *How do we ensure fairness and reliability across programmes?* **A:** *Standardization is built in. Every programme follows the same process, ensuring fairness, validity, and reliability. This consistency strengthens both institutional credibility and learner trust.*

▨ TECHNICAL CONCERNS

Q: *What if the system fails or data is lost?* **A:** *Outputs are version-controlled and backed up. Institutions retain full ownership of their material. The system is designed with resilience and traceability at its core.*

▧ STRATEGIC IMPACT

Q: *What's the real benefit for institutions?* **A:** *Audit readiness, global competitiveness, and educator empowerment. Institutions can deliver accredited, compliant programmes faster, with less risk, and with more focus on pedagogy.*

🎯 DELIVERY TIPS FOR Q&A

- **Listen fully** before answering — pause, nod, then respond.
- **Anchor answers** to key phrases: *"Months to minutes," "Compliance assured," "Empowerment delivered."*
- **Keep answers concise** — 2–3 sentences maximum.
- **Close each answer** with confidence: *"This is how we ensure both compliance and transformation."*

DRAFT MOU TEMPLATE

Memorandum of Understanding (MoU) Between [Institution Name] and ETDP AI Learning Design System

1. PURPOSE

This MoU establishes a collaborative pilot programme between **[Institution Name]** and the **ETDP AI Learning Design System**, aimed at accelerating accredited learning material development in alignment with **QCTO and SAQA standards** and the **RSA AI Strategy (2026)**.

2. OBJECTIVES

- **Pilot Implementation**: Deploy the ETDP AI system to generate accredited learning resources for selected programmes.
- **Compliance Assurance**: Ensure outputs meet QCTO and SAQA requirements.
- **Educator Empowerment**: Reduce administrative burden, enabling educators to focus on teaching.
- **Policy Alignment**: Demonstrate alignment with RSA AI Strategy pillars (inclusive growth, ethical governance, data sovereignty, human capital development).

3. ROLES & RESPONSIBILITIES

- **Institution:**
  - Provide access to curricula and programme specifications.
  - Facilitate educator engagement and feedback.
- **ETDP AI Team:**
  - Deliver system access, training, and technical support.
  - Ensure compliance and standardization of outputs.

4. DURATION

This MoU is valid for **[6 months]**, commencing on **[Start Date]**, with option to extend based on pilot outcomes.

5. EVALUATION & REPORTING

- **Pilot Review** conducted at 3 months.
- **Final Report** submitted at 6 months, including compliance metrics, educator feedback, and institutional impact.

6. CONFIDENTIALITY & DATA SOVEREIGNTY

Both parties agree to safeguard institutional data, ensuring compliance with South African data protection laws and the RSA AI Strategy's principle of **local data sovereignty**.

7. SIGNATURES

For [Institution Name] Name: _____ Title: _____ Date: _____

For ETDP AI Learning Design System Name: Dr Pierre Wepener Title: Technical Project Lead Date: _____

✨ This MoU template is concise, enforceable, and symbolic — it positions your pilot programmes as **policy-aligned partnerships** rather than mere trials.

PILOT EVALUATION FRAMEWORK

CORE METRICS

- **Compliance Accuracy** % of outputs fully aligned with QCTO and SAQA standards.
- **Time Efficiency** Average time to generate complete accredited material vs. traditional methods.
- **Educator Empowerment** Survey results on reduced administrative burden and increased teaching focus.
- **Learner Readiness** Feedback on clarity, accessibility, and usability of generated materials.
- **Institutional Impact** Audit readiness, risk reduction, and programme expansion potential.

EVALUATION METHODS

- **Baseline Comparison** Compare pilot outputs against existing manual curricula.
- **Educator Surveys** Structured questionnaires on workload, satisfaction, and empowerment.
- **Learner Feedback** Focus groups and surveys on material usability and learning outcomes.
- **Compliance Audit** Independent review of outputs against accreditation standards.
- **Performance Dashboard** Visual tracking of metrics (time saved, compliance %, satisfaction scores).

REPORTING TIMELINE

- **Month 1:** Initial educator feedback + compliance spot check.
- **Month 3:** Mid-pilot review — efficiency metrics + learner feedback.

- **Month 6:** Final report — comprehensive evaluation across all metrics.

▨ SUCCESS INDICATORS

- Compliance accuracy ≥ 95%.
- Time efficiency: material generated in < 40 minutes.
- Educator empowerment: ≥ 80% report reduced admin burden.
- Learner readiness: ≥ 75% positive feedback on usability.
- Institutional impact: ≥ 2 programmes audit-ready within pilot period.

✨ With this framework, you'll be able to **prove the system's value in hard numbers and lived experience** — exactly what stakeholders, funders, and policymakers need to see.

Would you like me to now prepare a **performance dashboard design** — a visual outline showing how these metrics can be tracked and presented in reports and presentations?

📊 PERFORMANCE DASHBOARD DESIGN

▧ DASHBOARD SECTIONS

1. **Compliance Accuracy**
   - Metric: % of outputs aligned with QCTO/SAQA standards
   - Visual: Gauge chart (target ≥ 95%)
2. **Time Efficiency**
   - Metric: Average time to generate material vs. traditional baseline
   - Visual: Bar chart (baseline vs. pilot)
3. **Educator Empowerment**
   - Metric: % of educators reporting reduced admin burden
   - Visual: Survey results pie chart (target ≥ 80%)
4. **Learner Readiness**
   - Metric: % of learners rating materials "clear and usable"
   - Visual: Satisfaction bar chart (target ≥ 75%)
5. **Institutional Impact**
   - Metric: # of programmes audit-ready within pilot period

- Visual: Progress tracker (target ≥ 2 programmes)

▨ DASHBOARD LAYOUT

- **Top Row:** Compliance gauge + time efficiency bar chart.
- **Middle Row:** Educator empowerment pie chart + learner readiness bar chart.
- **Bottom Row:** Institutional impact progress tracker.
- **Footer:** Key success indicators + RSA AI Strategy alignment badges.

▧ REPORTING FLOW

- **Monthly snapshots**: quick visual updates for stakeholders.
- **Mid-pilot review (Month 3)**: dashboard presented in roundtable.
- **Final report (Month 6)**: dashboard included in Minister briefing + media pack.

✨ This dashboard design ensures your pilot results are **visual, persuasive, and policy-aligned** — turning metrics into a compelling story of transformation.